Fusion Plasma Two-Fluid Treatment

Personal Notes

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Preface

The following are some personal notes I have written during my self-study of fusion and plasmas. I want to disclaim that I have not yet taken any courses that explain these topics in detail. I have only taken a brief introduction through the course "Plasmas and Electromagnetism" in Aix-Marseille Université, and the standard courses that are offered in a physics bachelor's degree.
The notes attempt to cover the two-fluid treatment of fusion plasmas, through the use of their particle, momentum and energy conservation equations (with state equations). The usual terms that make up these equations are derived and explained. In the end, a couple of results that can be obtained from this treatment are also found.
Using the results that are found for the bootstrap current, a simulation of the current in a toroidal plasma was made. First, a current is induced into the plasma via a central solenoid, using the ramp-up and flat-top phases. Then the bootstrap effect is added, which increases the total plasma current as time passes. Analyses are made for the total plasma current, and the fraction of the current that is generated as bootstrap is compared to the fraction that is induced by the central solenoid. The source code, results and discussion can be found at [5].
These notes are intended to build upon the foundation that was established during the making of my bachelor's thesis. They then require a certain level of previous knowledge to be fully understood. To get the full context of the content that will be discussed in this document, I may recommend my own thesis ([6]), or any of the textbooks that are there cited.
The main source I have used to study and write these notes is Freidberg's "Plasma Physics and Fusion Energy" [3], although Goldston's "Introduction to Plasma Physics" [4] was also very useful (specially for chapter 6). Chen's "Introduction to Plasma Physics and Controlled Fusion" [2] was used to make some clarifications, as well as the available notes for MIT's "Introduction to Plasma Physics I" and other cited sources.

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Chapter 1

Introduction

The description of a fusion plasma as a fluid starts with the derivation of "separate" and individual fluid equations, for the ion species and for the electrons. Normally, these are the only species that are taken into account, although technically other particles, like the α particles that result from fusion, need to be considered in certain situations. The need to have a set of equations for each species lies in the possible differences between their mass and charge, which makes them behave differently under the same stimulus.
These fluid equations are obtained by first analyzing the conservation of different quantities in the plasma, mainly the number of particles, their momentum and their energy.

However, this fluid treatment of plasma is not entirely correct. The utility of a fluid model relies on whether it can simplify the study of a given system, without making its description inaccurate. Part of the simplification requires that a system of N particles, and a total volume of V, can be divided into smaller elements of volume ΔV, with a high enough number of particles ΔN within them to make the statistical simplification of taking average quantities a worthwhile procedure.
On the other side, if in order to have a valid number of particles ΔN, the volume ΔV gets too big, then the spatial resolution of the system will be too poor, and the accuracy of the model will be unacceptably low.

To evaluate then whether this fluid treatment is a valid approach for fusion plasmas, we can compare it with a well known fluid that is indeed properly described through such physics. One of this fluids is air, at room temperature and atmospheric pressure. Even though air is some ten thousand times more dense than a fusion plasma, both systems can be divided into sub elements that both justify their statistical treatment, and also keep a good spacial resolution of the system.
But there is a big difference between the fluid elements in air and those of a plasma, which is the identity of the particles within that element. In air, molecules are constantly colliding with each other, and these collisions prevent the molecules from traveling very far from their initial position. If each molecule suffers a random collision with a mean time τ, then it will be unlikely that it will travel too far in any specific direction, and it

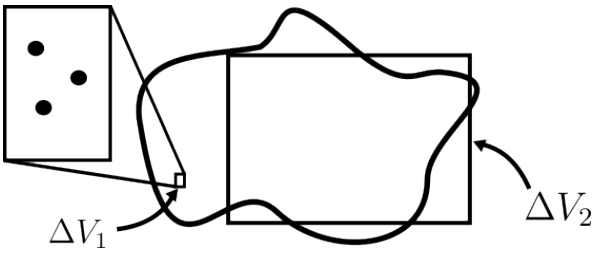


Figure 1.1: Diagram showing different invalid choices of fluid element size. ΔV_1 contains a very small number of particles, while ΔV_2 has a size comparable to the system.

will probably stay within a region of space comparable to the mean free path λ as time passes in the system. In air, the collisions between the molecules give coherence to the system, and in this way a fluid element of volume ΔV can almost be treated like a single particle with a mass equal to the total mass of the ΔN particles, and a velocity equal to their average.

In plasmas, however, this coherence is not complete. In fact, one of the main differences between a plasma and a regular gas is the fact that collisions between plasma particles are almost irrelevant when it comes to the dynamics of the system. Still, the Lorentz force can provide some means of spatial confinement for the plasma particles. When applying a magnetic field to the plasma, its particles will gyrate around the field lines in a plane perpendicular to them. This gyration will keep the particles from moving too far from their original position, to some extent, albeit only in the perpendicular direction to the field.
In the parallel direction, there is no force that will keep the particles confined to a small enough region of space. The parallel component of the velocity of a particle can vary within a wide spectrum, and then two particles that were initially beside each other will drift apart. Thus, in plasmas, while there may be coherence in the directions perpendicular to the magnetic field, there will none in the parallel direction. Then, the fluid description of a plasma will not be entirely justified, at least not in the direction parallel to the magnetic field.

Still, plasmas are often modeled as if they were complete fluids, and through these models a lot of behaviors can be described with a sufficient accuracy. This is because, in many situations, the parallel motion of the system has little importance in its overall behavior. In these cases (some of which will be treated near the end of this document), the inaccuracy of also treating the plasma as a fluid in its parallel motion has little effect on the obtained results. In other situations, this treatment yields inaccurate results, while still being qualitatively correct. Some other known plasma phenomena cannot be described in any way through a fluid treatment.

Even then, a lot of the properties of plasma can be studied and analyzed by treating

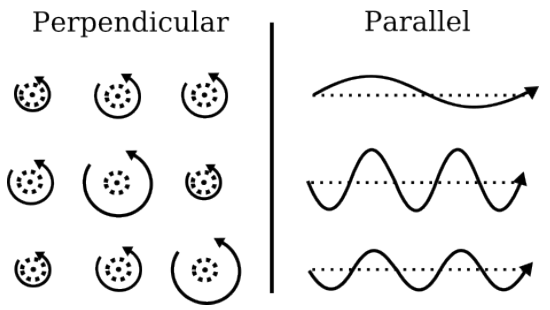


Figure 1.2: Diagram showing the differences between motion perpendicular and parallel to the field. The magnetic field is represented by the dotted lines, while the full ones are the trajectory of the plasma particles (in this case positively charged). Note the differences in the trajectories, as not all particles have the same velocities.

the system as a fluid, and this study is not completely irrelevant in the quest of better understanding the physics of fusion plasmas. The fluid treatment also has the added benefit of being the most conceptually simple, and the easiest to understand intuitively, as opposed to kinetic theory.

Quantities of Interest

When using fluid models to describe a system, the main quantities that are used for this description are the particle number density n of the fluid, and its average velocity $\vec{u}$. Remember that these quantities are given locally for a specific element of the total system, and they are also given individually for each species in the plasma. Then, if an element of volume ΔV contains a number N of particles inside it, the particle density n will be given like:

$$n = \frac{N}{\Delta V}$$

The average of macroscopic velocity $\vec{u}$ will then just be the (regular) average velocity of all the particles:

$$\vec{u} = \frac{\sum\limits_i \vec{v}_i}{N}$$

Where i represents each particle. In other words, $\vec{u}$ will be the sum of all the particles' velocities, divided by the number of particles.

As we shall see, the fluid equations relate these two quantities with each other, as well as with other quantities. For the conservation of momentum, the force that acts on the

fluid element will be important. Seeing how the plasma is made up of charged particles, the force $\vec{F}$ that will act on the fluid element will be the Lorentz force:

$$\vec{F} = q\left(\vec{E} + \vec{u} \times \vec{B}\right)$$

So then, the electromagnetic fields $\vec{E}$ and $\vec{B}$ that act on the fluid element will also be quantities of interest.
For the conservation of energy, the energy sources and sinks of the plasma will also be of importance. This will concern quantities like the heating power input into the plasma, as well as the radiation power output by the plasma. These quantities will require their own sections to be properly explained.

One last precision needs to be made, about the reference systems that will be used when deriving the fluid equations. There are two frames that would make sense to use. The first is the one that moves with the fluid element, which is also called the Lagrangian coordinate system. It is simpler to imagine the system in this way, and thus it is easier to first write the equations in this form. The other frame is the one of the laboratory, called the Eulerian system. This one is more practical to use, in the sense that this is the real frame at which plasmas are studied experimentally.

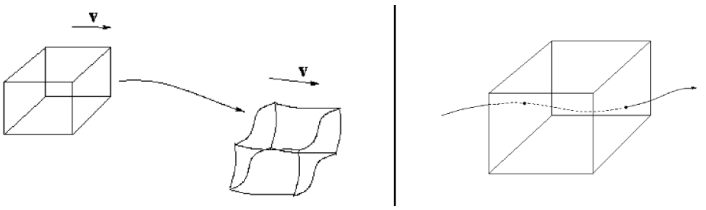


Figure 1.3: Diagram showing the differences between the reference frames. To the left, the Lagrangian frame moves with the fluid, while to the right, the Eulerian frame stays put while the fluid flows through it. Taken from [7].

The rate of change of a quantity Q of the fluid is given differently in each system. If we denote the Lagrangian system with $\mathcal{L}$, and the Eulerian system with $\mathcal{E}$, then the Lagrangian time derivative will be given in the Eulerian system like:

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\mathcal{L}} = \left(\frac{\partial Q}{\partial t} + \left(\vec{u} \cdot \vec{\nabla}\right) Q\right)_{\mathcal{E}}$$

The Lagrangian derivative is also commonly called the convective, or material, derivative. A common procedure could be to write down the dynamics of the quantity Q in the more intuitive Lagrangian form, to then convert it to the more practical Eulerian form.

Chapter 2

Conservation of Particles

The conservation of the number of particles starts with the assumption that this number remains constant for each species, at all times. This means that electrons and ions will continue to be separate particles, and that neither of them will be created or "destroyed" through any processes. Both of these assumptions are valid for fusion plasmas, as their high temperatures will keep atoms ionized, and will prevent ions and electrons from recombining.
The fusion part of the "fusion plasma" is a little counter-intuitive, as fusion reactions will change the identities of the ions, and will also introduce other types of particles into the plasma. However, the frequency of these fusion reactions is low, and so the loss of particles through this effect can be ignored. As for the fusion products that stay within the plasma, if they eventually grow so much in quantity that they need to be taken into account, this can be corrected by just adding another set of fluid equations for those species.

Then, consider a fluid element, of volume ΔV and with a particle density n. With this, the condition for the conservation of the number of particles may be written like:

$$\begin{aligned} \frac{\mathrm{d}N}{\mathrm{d}t} &= 0 \\ &= \frac{\mathrm{d}n}{\mathrm{d}t}\Delta V + n\frac{\mathrm{d}\Delta V}{\mathrm{d}t} \end{aligned} \tag{2.0.1}$$

Where the differentiation rule for a product was used. Furthermore, if the volume ΔV can also be written like equal to $\Delta x \Delta y \Delta z$, then its derivative will be written like:

$$\frac{\mathrm{d}\Delta V}{\mathrm{d}t} = \frac{\mathrm{d}\Delta x}{\mathrm{d}t}\Delta y \Delta z + \Delta x \frac{\mathrm{d}\Delta y}{\mathrm{d}t}\Delta z + \Delta x \Delta y \frac{\mathrm{d}\Delta z}{\mathrm{d}t}$$

What Equation 2.0.1 tells us is that both the particle density n and the volume ΔV can change through time, but they have to do so in equal and opposite amounts so that the total number of particles inside of the element stays constant.

To continue, we then need to find a way of expressing the change in the dimensions of the volume element as time changes. To do this, let us first consider only one side, for

example the one along the x axis. Let's consider a fluid element, with boundaries along

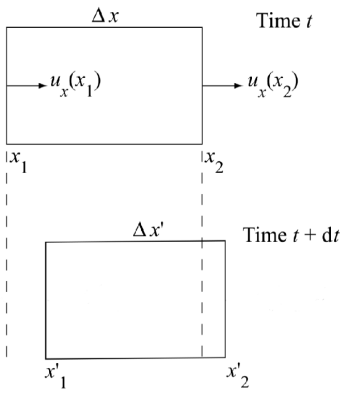


Figure 2.1: Diagram showing how the thickness of a fluid element may change as time passes. Adapted from [3].

the x axis in the points x_1 and x_2, which would give it a corresponding thickness Δx of $x_2 - x_1$. Each boundary moves with a velocity $u_x(x_1)$ and $u_x(x_2)$ respectively. Then, after a time $\mathrm{d}t$ passes, each boundary will have moved a distance $u_x(x)\mathrm{d}t$, and so the new thickness $\Delta x'$ of the fluid element along the x axis will be given like:

$$\begin{aligned}\Delta x' &= x_2 + u_x(x_2)\mathrm{d}t - (x_1 + u_x(x_1)\mathrm{d}t) \\ &= \Delta x + (u_x(x_2) - u_x(x_1))\,\mathrm{d}t\end{aligned}$$

Now, this distance Δx is supposed to be very small (remembering the resolution requirements for the fluid model to work), so we can try to expand this expression around $\Delta x \approxeq 0$ ($x_1 \approxeq x_2$). We would then have:

$$\begin{aligned}\Delta x' &\approxeq \Delta x + (u_x(x_2) - u_x(x_1))\,\mathrm{d}t \\ &\quad + \left(\frac{\partial x_1}{\partial x} - \frac{\partial x_2}{\partial x}\right) + \left(\frac{\partial u_x(x_2)}{\partial x} - \frac{\partial u_x(x_1)}{\partial x}\right)(x_2 - x_1)\,\mathrm{d}t \\ &\quad + \dots \\ &\approxeq \Delta x + 0 + (1-1) + \left(\frac{\partial\,(u_x(x_2) - u_x(x_1))}{\partial x}\right)\Delta x\ \mathrm{d}t \\ &\approxeq \Delta x + \frac{\partial u_x}{\partial x}\Delta x\ \mathrm{d}t\end{aligned}$$

This is the expression for the new thickness $\Delta x'$. We want to find an expression for the change in thickness as time passes. This could be obtained by calculating the difference

between $\Delta x'$ and Δx, and dividing over the change in time:

$$\begin{aligned}\frac{\mathrm{d}\Delta x}{\mathrm{d}t} &= \frac{\Delta x' - \Delta x}{\mathrm{d}t} \\ &= \frac{\partial u_x}{\partial x}\Delta x\end{aligned}$$

Following this same procedure for the remaining dimensions, we would obtain the same results. Then, we would end up writing the change in volume with time as:

$$\begin{aligned}\frac{\mathrm{d}\Delta V}{\mathrm{d}t} &= \Delta x\Delta y\Delta z\left(\frac{\partial u_x}{\partial x}\right) + \Delta x\Delta y\Delta z\left(\frac{\partial u_y}{\partial y}\right) + \Delta x\Delta y\Delta z\left(\frac{\partial u_z}{\partial z}\right) \\ &= \Delta V\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}\right) \\ &= \Delta V\left(\vec{\nabla}\cdot\vec{u}\right)\end{aligned}$$

And then the conservation of the number of particles would be written like:

$$\frac{\mathrm{d}n}{\mathrm{d}t}\Delta V + n\Delta V\left(\vec{\nabla}\cdot\vec{u}\right) = 0$$

$$\frac{\mathrm{d}n}{\mathrm{d}t} + n\left(\vec{\nabla}\cdot\vec{u}\right) = 0$$

Finally, transforming the Lagrangian time derivative into its equivalent in the Eulerian lab frame, we would obtain:

$$\begin{aligned}\left(\frac{\partial n}{\partial t} + \left(\vec{u}\cdot\vec{\nabla}\right)n\right) + n\left(\vec{\nabla}\cdot\vec{u}\right) &= 0 \\ \frac{\partial n}{\partial t} + \vec{u}\cdot\left(\vec{\nabla}n\right) + n\left(\vec{\nabla}\cdot\vec{u}\right) &= \\ \boxed{\frac{\partial n}{\partial t} + \vec{\nabla}\cdot(n\vec{u}) = 0} \qquad (2.0.2)\end{aligned}$$

Where, once again, the product rule was used.

This last expression can be recognized as the usual form of the continuity equation, in this case for the number of particles in the fluid element.
What it tells us is that the change in the number of particles inside of said fluid element will only be due to the flow of particles into or out of it. This flow then comes from, or goes into, the adjacent fluid elements. Thus, the total number of particles in the whole system remains constant.

By multiplying Equation 2.0.2 by the mass of each species, we could get an expression for the conservation of mass, and by multiplying it by their charge, we could get the conservation of charge in the system.

Chapter 3

Conservation of Momentum

The conservation of momentum in a fluid is given by Newton's second law, which says that the change in the momentum of a body is equal to to the force that acts upon it [1].

The momentum of the fluid element will be given by the total mass of the particles inside the fluid element, times their average velocity.
The principal forces that will be considered to act on a given element will be the electromagnetic force, the pressure gradient force and the collisional force. There are other forces which technically also act on the elements, like gravity and the viscous force, but their effects are negligible in a fluid such as a fusion plasma.

3.1 Change in Momentum

The momentum of a fluid element of volume ΔV and particle density n, with particles of mass m, will be given like:

$$\begin{aligned} \vec{P} &= M\vec{u} \\ &= (\Delta V n m)\,\vec{u} \end{aligned}$$

[1] For Newton's second law, this "body" technically needs to be a point particle, or an object that can be simplified as one. Even though we are actually considering a collection of multiple particles, in the fluid treatment an element is treated as a "super particle", with parameters equal to the average of all of the particles inside the element. Then, Newton's second law can be properly applied when studying the dynamics of fluids.

Where m is the mass of a single particle. So then, the change through time for the momentum of the element will be given like:

$$\frac{\mathrm{d}\vec{P}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\Delta V n m\right)\vec{u}$$

$$= \boxed{\Delta V n m \left(\frac{\partial \vec{u}}{\partial t} + \left(\vec{u}\cdot\vec{\nabla}\right)\vec{u}\right)}$$

Where the particle density can be treated as a constant if the plasma is homogeneous, which is a fair and usual assumption.

3.2 Electromagnetic Force

The electromagnetic force that acts on the fluid element will be given as the Lorentz force. For a similar fluid element of volume ΔV and particle density n, with singular particles of charge q, this would be given by the expression:

$$\vec{F} = Q\left(\vec{E} + \vec{u}\times\vec{B}\right)$$

$$= \boxed{\Delta V n q \left(\vec{E} + \vec{u}\times\vec{B}\right)}$$

The fields $\vec{E}$ and $\vec{B}$ are the long range electric and magnetic fields that act on the plasma element. The term "long range" includes all fields, excepting the ones that mediate the "short range" Coulomb collisions between plasma particles. This means that the whole range of electromagnetic interactions is treated in this momentum conservation model through the electromagnetic force term and the collision term.

Maxwell's Equations

The long range field is composed of both external fields, produced by the machine that generated the plasma, and fields internal to the plasma itself.
The force term component due to the external machine fields will most likely be able to be calculated, given the likelihood of knowing the fields that are produced by the machine.
The internal, "self consistent", fields generated by the plasma will be calculated through Maxwell's equations. To do this, we will need the charge density ρ of the plasma, as well as its current density $\vec{J}$. These two quantities will also be related to each other through the continuity equation that can be obtained from the conservation relation given in Chapter 2.

For the charge density, if the plasma has multiple species χ, with particles of charge

q_χ, then the total charge density of the plasma will be given like:

$$\rho = \sum_\chi q_\chi n_\chi$$

Where n_χ is the particle density of each species.
In a very similar way, the current density will be obtained like:

$$\begin{aligned}\vec{J} &= \sum_\chi \rho \vec{u}_\chi \\ &= \sum_\chi q_\chi n_\chi \vec{u}_\chi\end{aligned}$$

Where a summation must also be carried through the average velocity of each species, as this quantity may be different for each of them.

Now that ρ and $\vec{J}$ have been obtained, the electromagnetic fields generated by the plasma can be calculated with Maxwell's equations, which would take the form:

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= \frac{\sum_\chi q_\chi n_\chi}{\varepsilon_0} \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{B} &= \mu_0 \left(\sum_\chi q_\chi n_\chi \vec{u}_\chi + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right)\end{aligned}$$

3.3 Pressure Gradient Force

This force is due to the flow of particles through the boundaries of the fluid element. As particles enter or exit the element, the element's overall velocity and momentum will change depending on the direction of the particle flow. So the effect of this flow of particles can be quantified through a force, which is the pressure gradient force.
This flow of particles is due to small variations in the velocities of the particles. The velocity $\vec{u}$ of the fluid element is defined as the average velocity of all the particles inside of the element, however, each particle has a velocity that is slightly different from this average value.
The velocity $\vec{v}$ of a particle in the element may be given as equal to $\vec{u} + \vec{w}$, where $\vec{w}$ is the small deviation due to the thermal motions of the particle. In fusion plasmas, this thermal velocity $\vec{w}$ follows a Maxwell-Boltzmann distribution.

Let's first examine the particle flow in the $\hat{x}$ direction. If the element has a volume $\Delta V = \Delta x \Delta y \Delta z$, then for an element which is centered at a given x coordinate, the

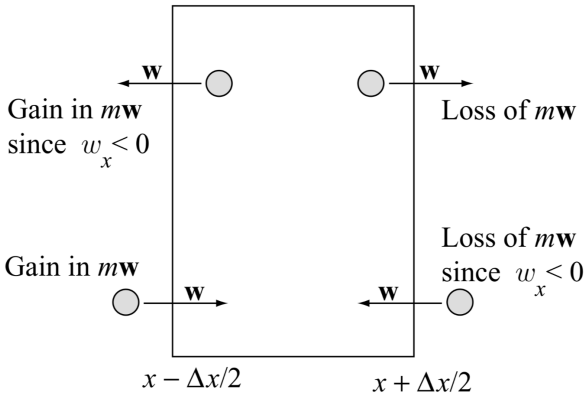


Figure 3.1: Diagram showing the possible particle flows through the fluid element's boundaries (in the $\hat{x}$ direction), and their corresponding momentum changes. Taken from [3].

boundaries through which the particles will flow are placed on $x \pm \Delta x/2$, as can be seen in Figure 3.1.
If this fluid element is moving in the positive $\hat{x}$ direction ($u_x > 0$), then the positive momentum direction will be defined in the $+\hat{x}$ direction, and each particle flow will change the element's momentum like:

- A particle leaving though the left boundary will increase the element's overall momentum, as a particle with negative momentum will leave the element.
- A particle leaving through the right boundary will decrease the overall momentum of the element, because it will lose a particle with positive momentum.
- A particle entering through the left boundary will increase the overall momentum, as the element will gain a particle with positive momentum.
- A particle entering through the right boundary will decrease the overall momentum of the element, as it will gain a particle with negative momentum.

Consider first the particles that leave through the right boundary. First, we can get the differential $\mathrm{d}n$ of the number density of particles using the Maxwellian distribution f_{M} (function of the position $\vec{r}$, the thermal velocity $\vec{w}$ and the time t). Multiplying by a velocity range $\mathrm{d}\vec{w}$, we can obtain the particle range $\mathrm{d}n$, and evaluating at $x + \Delta x/2$, the particle density range is obtained at the right boundary of the fluid element.

To find the particle flow through the boundary, the particle density $\mathrm{d}n_{\mathrm{R}}$ ("R" for right boundary) can be multiplied by the area $\Delta y \Delta z$ of the boundary, times the speed w_x of the particles that cross that boundary. This would give the flow of particles that cross the right boundary of the fluid element, with velocities within the range $\mathrm{d}\vec{w}$.
Then, to find the total flow $I_{\mathrm{R}}^{\mathrm{O}}$ ("O" for out of the element) of particles that leave though

the right boundary, the differential would have to be integrated over the whole range of positive particle velocities. This is:

$$\begin{aligned} I_{\mathrm{R}}^{\mathrm{O}} &= \int \mathrm{d}n_{\mathrm{R}} w_x \Delta y \Delta z \\ &= \Delta y \Delta z \left(\int_0^{\infty} f_{\mathrm{M}} w_x \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \end{aligned} \tag{3.3.1}$$

This expression would give the number of particles per unit of time that exit the fluid element through its right boundary.
To find an expression for the change in the element's momentum, we will first only consider the change in the $\hat{x}$ directed momentum. Then, each particle that crosses the boundary would have a momentum $mw_x\hat{x}$, and multiplying this quantity times the flow of particles through the boundary would give the change of momentum per unit time (force $\vec{F}_{\mathrm{R}}^{\mathrm{O}}$) for the fluid element (which would be negative, for a loss of momentum). This would be:

$$\begin{aligned} \vec{F}_{\mathrm{R}}^{\mathrm{O}} &= -I_{\mathrm{R}}^{\mathrm{O}} \left(m w_x \hat{x} \right) \\ &= -m \Delta y \Delta z \left(\int_0^{\infty} f_{\mathrm{M}} w_x^2 \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \hat{x} \end{aligned}$$

A very similar expression would arise for an analysis of the particles that enter through the right boundary. In fact, it would be the same expression, just with integration limits from $-\infty$ to 0, to account for the negative velocities the particles require to enter through the right side:

$$\vec{F}_{\mathrm{R}}^{\mathrm{I}} = -m \Delta y \Delta z \left(\int_{-\infty}^{0} f_{\mathrm{M}} w_x^2 \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \hat{x}$$

Notice also the negative sign, because the particles that enter through the right boundary also decrease the momentum of the element.

The total force $\vec{F}_{\mathrm{R}}$ due to the flow of particles flowing through the right boundary would be the addition of both integrals, which would give the corresponding integral though the entire particle velocity range:

$$\vec{F}_{\mathrm{R}} = -m \Delta y \Delta z \left(\int_{-\infty}^{\infty} f_{\mathrm{M}} w_x^2 \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \hat{x} \tag{3.3.2}$$

To simplify this expression, it could be noticed that this integral has a similar form to that of the integrals that define average quantities. For an arbitrary quantity Q, its average

over the (velocity part of the) Maxwellian distribution f_{M} would be given like:

$$\begin{aligned}\langle Q \rangle &= \frac{\int_{-\infty}^{\infty} Q f_{\mathrm{M}} \, \mathrm{d}\vec{w}}{\int_{-\infty}^{\infty} f_{\mathrm{M}} \, \mathrm{d}\vec{w}} \\ &= \frac{\int_{-\infty}^{\infty} Q f_{\mathrm{M}} \, \mathrm{d}\vec{w}}{n}\end{aligned}$$

Using this, Equation 3.3.2 could be written like:

$$\vec{F}_{\mathrm{R}} = -m\Delta y \Delta z \left(n \left\langle w_x^2 \right\rangle \right) \Big|_{x+\Delta x/2} \hat{x}$$

And here can be noticed the quantity $mn \left\langle w_x^2 \right\rangle$, which is the definition of the pressure p (as seen in [2]) of the fluid element [2].
Then, the force due to the flow of particles through the right boundary will be finally written like:

$$\vec{F}_{\mathrm{R}} = -p \Big|_{x+\Delta x/2} \Delta y \Delta z \hat{x}$$

Which is consistent with the definition of pressure being force over area.

The same process can be repeated for the left boundary, changing the evaluation point to $x - \Delta x/2$. This would give the same expression, changing the sign of the force, seeing how any flow of particles through the left boundary increases the overall momentum of the fluid element.
Then, the total force $\vec{F}$ due to the flow of particles in the $\hat{x}$ direction would be given like:

$$\vec{F}_x = \Delta y \Delta z \left(p \Big|_{x-\Delta x/2} - p \Big|_{x+\Delta x/2} \right)$$

Here once again we can do a Taylor expansion for very small values of Δx, which would give:

$$\begin{aligned}\vec{F}_x &\approxeq \Delta y \Delta z \left(p \Big|_{x-\Delta x/2} - p \Big|_{x+\Delta x/2} + \Delta x \left(\frac{\partial p}{\partial x} \Big|_{x-\Delta x/2} - \frac{\partial p}{\partial x} \Big|_{x+\Delta x/2} \right) \right) \\ &\approxeq \Delta y \Delta z \left(0 - \Delta x \left(\frac{\partial p}{\partial x} \Big|_{x+\Delta x/2} - \frac{\partial p}{\partial x} \Big|_{x-\Delta x/2} \right) \right) \\ &\approxeq -\Delta V \frac{\partial p}{\partial x}\end{aligned}$$

[2] It is actually a special case of the definition of the pressure tensor P, which will be discussed further on.

Where the minus sign comes from the fact that the derivatives are evaluated in an inverse order [3].
The above expression gives the force in the $\hat{x}$ direction. Similar expressions would be found repeating this procedure for the other corresponding directions. Adding them all together, it would give:

$$\begin{aligned}\vec{F} &= -\Delta V\left(\frac{\partial p}{\partial x}+\frac{\partial p}{\partial y}+\frac{\partial p}{\partial z}\right)\\ &\approx \boxed{-\Delta V\vec{\nabla}p}\end{aligned}$$

An so it can be seen why this force, which initially seemed to be unrelated to pressure, is called the pressure gradient force.

Plasma Pressure (Stress Tensor)

To find an expression for the pressure of the plasma, the integral that defines $mn\left\langle w_x^2\right\rangle$ can be calculated for the Maxwellian distribution f_{M}. Writing it explicitly, this is:

$$mn\left\langle w_x^2\right\rangle = mn\left(\frac{m}{2\pi k_B T}\right)^{3/2}\int_{-\infty}^{\infty} w_x^2\exp\left(-\frac{m\left(w_x^2+w_y^2+w_z^2\right)}{2k_BT}\right)\,\mathrm{d}w_x\mathrm{d}w_y\mathrm{d}w_z$$

Where it is important to remember that the n besides the integral comes from the normalization constant of the Maxwell-Boltzmann distribution (it is not the same n that is multiplying the average $\left\langle w_x^2\right\rangle$.)

The integral over w_x is a special integral, while the other two integrals have the form of the Gaussian integral. All of them have well known answers (which can be retrieved with a standard special integral table, or something like [9]), so replacing their values we obtain:

$$\begin{aligned}mn\left\langle w_x^2\right\rangle &= mn\left(\frac{m}{2\pi k_BT}\right)^{3/2}\left(\frac{\sqrt{\pi}}{2}\left(\frac{2k_BT}{m}\right)^{3/2}\right)\left(\sqrt{\pi}\left(\frac{2k_BT}{m}\right)^{1/2}\right)\left(\sqrt{\pi}\left(\frac{2k_BT}{m}\right)^{1/2}\right)\\ &= mn\left(\frac{m}{2\pi k_BT}\right)^{3/2}\left(\frac{2\pi k_BT}{m}\right)^{3/2}\frac{k_BT}{m}\\ &= \boxed{nk_BT}\end{aligned} \tag{3.3.3}$$

This is the usual expression for the pressure of a plasma.

[3] Normally, derivatives are defined as the function in one point, minus the function at another point to the left of the first point. In this case, it was the opposite, the function p was evaluated first at a point $x-\Delta x/2$, minus the same function evaluated at a point $x+\Delta x/2$ to the right of the first point. So to write the whole expression as the derivative of p, the minus sign is needed to invert the subtraction to the usual order.

However, it must be noticed that this calculation is only a specific case of a more general one. First, in this case, the momentum change was calculated in the $\hat{x}$ direction, and it was assumed that it could only change due to the $\hat{x}$ component of the particles that flow through the boundaries. These two $\hat{x}$ components are what resulted in the expression $mn\langle w_x w_x\rangle = mn\langle w_x^2\rangle$.
However, a flow of particles through these same boundaries (due to the $\hat{x}$ component of the momentum of the particles) can also cause a change in momentum in other directions. The particles that flow through the boundaries can also have momentum in the $\hat{y}$ direction, and so under certain conditions the flow of particles in the $\hat{x}$ direction can also affect the $\hat{y}$ component of the element's momentum. This change would be calculated through a term proportional to $\langle w_x w_y\rangle$.

This is also true for the other directions, and the other combinations. Then, to take into account all of the possible flows of particles and their corresponding change in momentum of the fluid element, the pressure can no longer be given as a simple number, but rather as a tensor P, called the stress tensor, with components $p_{ij} = mn\langle w_i w_j\rangle$ that represent the different flows of particles:

$$P = \begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{yx} & p_{yy} & p_{yz} \\ p_{zx} & p_{zy} & p_{zz} \end{pmatrix}$$

The first index of the component represents the direction of the flow of the particles (the boundary plane will be normal to the direction in this index), and the second component represents the direction in which the momentum of the fluid element will be changed.
The components in the diagonals are called the normal stresses, which were the only ones that were calculated previously. The other mixed components are called shear stresses, which are also a possibility in fusion plasmas.

Then, a more general form of writing the pressure gradient force would be to replace the pressure p for the stress tensor P:

$$\boxed{\vec{F} = -\Delta V \vec{\nabla} P}$$

Another assumption was that the plasma fluid was isotropic, so the result that was obtained for the $\hat{x}$ direction was also extended to the other directions.
However, it can be remembered that this is not true at all for plasmas under the influence of a magnetic field $\vec{B}$, which behave differently in the directions perpendicular and parallel to the field. Then, a more general derivation of the stress tensor should not assume that the normal stresses are all equal to the usual plasma pressure, that $p_{xx} = p_{yy} = p_{zz} = nk_B T$.

Consider for example a plasma fluid, with no shear stresses, under the influence of a magnetic field in the $\hat{z}$ direction. In this case, the plasma would have a different motion in the $\hat{z}$ direction that in the $\hat{x}$ and $\hat{y}$ directions.
This would be represented with different pressures in each direction, which would in turn

mean different temperatures in each direction:

$$p_{\parallel} = nk_B T_{\parallel}$$
$$p_{\perp} = nk_B T_{\perp}$$

The stress tensor for this plasma would be written like:

$$P = \begin{pmatrix} p_{\perp} & 0 & 0 \\ 0 & p_{\perp} & 0 \\ 0 & 0 & p_{\parallel} \end{pmatrix}$$

3.4 Collision Force

There is one more force, due to the collisions inside of the fluid element. These collisions are not physical collisions, as the particles that make up a plasma are electrically charged. Then, the particles will be electrically attracted or repulsed way before they can come into physical contact, but they will still be affected as if they had physically collided.
This means these "collisions" are mediated through the electromagnetic potential, and are denominated Coulomb collisions. These collisions are the ones that take into account the short range electromagnetic interactions in the plasma.
Another thing which must also be made clear is that these don't take into account collisions between particles of the same species, because they will all make part of the same thermal distribution, and collisions between themselves will not change the overall momentum of the species in any way.
The collisions that do change the overall momentum of a group of particles are the collisions between different species. Consider again the fluid element of volume ΔV, where there are particles of species χ and ζ. Within that element, the species χ will have an overall momentum $\vec{p}_{\chi} = \Delta V n_{\chi} m_{\chi} \vec{u}_{\chi}$, and ζ will have a momentum $\vec{p}_{\zeta} = \Delta V n_{\zeta} m_{\zeta} \vec{u}_{\zeta}$.
If $\vec{p}_{\chi} \neq \vec{p}_{\zeta}$, then collisions between the particles of the two species will change their momentums, and will thus produce a force.
If the momentum of both species were the same, then the average over all collisions between their particles would not produce any change in their overall momentum, as there would not be any reason as to why one of them should gain momentum while the other one loses.

The principal repercussion that these Coulomb collisions have is that they will bring the different species of the plasma into thermal equilibrium. That is, they will make all the species to tend toward the same temperature value.
Even though we are currently discussing only momentum, and not energy, the change in the energy of the two species is also a necessary result from Coulomb collisions. Due to the fact that the electromagnetic force is conservative, the collisions between particles will be elastic.

Then, to be able to write an expression for the collision force in the plasma fluid, we first need to find an expression for the final macroscopic velocities $\vec{u}^{\mathrm{f}}$ of each species, in terms of their initial velocities $\vec{u}^{0}$.

The calculation of these expressions starts with the conservation of momentum and energy before and after the collision. Temporarily using $M = \Delta V n m$, this would be:

$$M_\chi \vec{u}^0_\chi + M_\zeta \vec{u}^0_\zeta = M_\chi \vec{u}^{\mathrm{f}}_\chi + M_\zeta \vec{u}^{\mathrm{f}}_\zeta$$

$$\frac{1}{2} M_\chi {u^0_\chi}^2 + \frac{1}{2} M_\zeta {u^0_\zeta}^2 = \frac{1}{2} M_\chi {u^{\mathrm{f}}_\chi}^2 + \frac{1}{2} M_\zeta {u^{\mathrm{f}}_\zeta}^2$$

From the first equation we can get:

$$\vec{u}^{\mathrm{f}}_\chi = \vec{u}^0_\chi + \frac{M_\zeta}{M_\chi} \left(\vec{u}^0_\zeta - \vec{u}^{\mathrm{f}}_\zeta \right) \tag{3.4.1}$$

And replacing this into the second equation, it would give:

$$\begin{aligned} M_\chi {u^0_\chi}^2 + M_\zeta {u^0_\zeta}^2 &= M_\chi \left(\vec{u}^0_\chi + \frac{M_\chi}{M_\zeta} \left(\vec{u}^0_\zeta - \vec{u}^{\mathrm{f}}_\zeta \right) \right)^2 + M_\zeta {u^{\mathrm{f}}_\zeta}^2 \\ &= M_\chi {u^0_\chi}^2 + 2 M_\zeta \vec{u}^0_\chi \cdot \left(\vec{u}^0_\zeta - \vec{u}^{\mathrm{f}}_\zeta \right) \\ &+ \frac{M_\zeta^2}{M_\chi} \left({u^0_\zeta}^2 - 2 \vec{u}^0_\zeta \cdot \vec{u}^{\mathrm{f}}_\zeta + {u^{\mathrm{f}}_\zeta}^2 \right) + M_\zeta {u^{\mathrm{f}}_\zeta}^2 \end{aligned}$$

Simplifying and putting all the unknowns to one side of the equation, this would give us:

$$M_\zeta {u^{\mathrm{f}}_\zeta}^2 + \frac{M_\zeta^2}{M_\chi} \left({u^{\mathrm{f}}_\zeta}^2 - 2 \vec{u}^0_\zeta \cdot \vec{u}^{\mathrm{f}}_\zeta \right) - 2 M_\zeta \vec{u}^0_\chi \cdot \vec{u}^{\mathrm{f}}_\zeta = M_\zeta {u^0_\zeta}^2 - 2 M_\zeta \vec{u}^0_\zeta \cdot \vec{u}^0_\zeta - \frac{M_\zeta^2}{M_\chi} {u^0_\zeta}^2$$

Now, if we multiply everything by M_χ / M_ζ, and we factorize, this can be written like:

$$ {u^{\mathrm{f}}_\zeta}^2 \left(M_\chi + M_\zeta \right) - 2 \vec{u}^{\mathrm{f}}_\zeta \cdot \left(M_\zeta \vec{u}^0_\zeta + M_\chi \vec{u}^0_\chi \right) = {u^0_\zeta}^2 \left(M_\chi - M_\zeta \right) - 2 M_\chi \vec{u}^0_\chi \cdot \vec{u}^0_\zeta$$

We can then divide everything over $M_\chi + M_\zeta$, and it would give:

$$ {u^{\mathrm{f}}_\zeta}^2 + \left(\frac{M_\zeta \vec{u}^0_\zeta + M_\chi \vec{u}^0_\chi}{M_\chi + M_\zeta} \right) \cdot \vec{u}^{\mathrm{f}}_\zeta = \left(\frac{M_\chi - M_\zeta}{M_\chi + M_\zeta} \right) {u^0_\zeta}^2 - 2 \left(\frac{M_\chi}{M_\chi + M_\zeta} \right) \vec{u}^0_\chi \cdot \vec{u}^0_\zeta$$

To the left we can notice a similar form to that of a binomial expansion, so if we complete the square, all of this can be written like:

$$\begin{aligned} &{u^{\mathrm{f}}_\zeta}^2 + \left(\frac{M_\zeta \vec{u}^0_\zeta + M_\chi \vec{u}^0_\chi}{M_\chi + M_\zeta} \right) \cdot \vec{u}^{\mathrm{f}}_\zeta + \left(\frac{M_\zeta \vec{u}^0_\zeta + M_\chi \vec{u}^0_\chi}{M_\chi + M_\zeta} \right)^2 \\ = &\left(\frac{M_\chi - M_\zeta}{M_\chi + M_\zeta} \right) {u^0_\zeta}^2 - 2 \left(\frac{M_\chi}{M_\chi + M_\zeta} \right) \vec{u}^0_\chi \cdot \vec{u}^0_\zeta + \left(\frac{M_\zeta \vec{u}^0_\zeta + M_\chi \vec{u}^0_\chi}{M_\chi + M_\zeta} \right)^2 \end{aligned}$$

$$\left(\vec{u}_\zeta^{\,\mathrm{f}} - \left(\frac{M_\chi \vec{u}_\chi^0 + M_\zeta \vec{u}_\zeta^0}{M_\chi + M_\zeta}\right)\right)^2 = \left(\frac{M_\chi - M_\zeta}{M_\chi + M_\zeta}\right) {u_\zeta^0}^2 - 2\left(\frac{M_\chi}{M_\chi + M_\zeta}\right) \vec{u}_\chi^0 \cdot \vec{u}_\zeta^0 + \frac{M_\zeta^2 {u_\zeta^0}^2 - 2M_\zeta M_\chi \vec{u}_\zeta^0 \cdot \vec{u}_\chi^0 + M_\chi^2 {u_\chi^0}^2}{\left(M_\chi + M_\zeta\right)^2}$$
$$= \frac{M_\chi^2 {u_\zeta^0}^2 - 2M_\chi^2 \vec{u}_\zeta^0 \cdot \vec{u}_\chi^0 + M_\chi^2 {u_\chi^0}^2}{\left(M_\chi + M_\zeta\right)^2}$$
$$= \frac{M_\chi^2 \left(\vec{u}_\zeta^0 - \vec{u}_\chi^0\right)^2}{\left(M_\chi + M_\zeta\right)^2}$$

And then we can finally find an expression for $\vec{u}_\zeta^{\,\mathrm{f}}$ as:

$$\vec{u}_\zeta^{\,\mathrm{f}} = \pm \frac{M_\chi \left(\vec{u}_\zeta^0 - \vec{u}_\chi^0\right)}{M_\chi + M_\zeta} + \frac{M_\chi + M_\zeta \vec{u}_\zeta^0}{M_\chi + M_\zeta}$$

Which would give the solutions:

$$\vec{u}_\zeta^{\,\mathrm{f}} = \vec{u}_\zeta^0,\ \frac{2M_\chi}{M_\chi + M_\zeta}\vec{u}_\chi^0 + \frac{M_\zeta - M_\chi}{M_\chi + M_\zeta}\vec{u}_\zeta^0 \tag{3.4.2}$$

And we can replace these solutions back into Equation 3.4.1 to get the corresponding velocities $\vec{u}_\chi^{\,\mathrm{f}}$ as:

$$\vec{u}_\chi^{\,\mathrm{f}} = \vec{u}_\chi^0,\ \frac{2M_\zeta}{M_\chi + M_\zeta}\vec{u}_\zeta^0 + \frac{M_\chi - M_\zeta}{M_\chi + M_\zeta}\vec{u}_\chi^0 \tag{3.4.3}$$

Looking at the obtained results, we could ignore the first of each, as they represent the case where the velocities don't change, and this does not make sense if we are considering the case where the particles are colliding [4].
Using the other solutions, we can write the change in velocity for each species, and also the change in momentum like:

$$\Delta\vec{u}_\chi = \vec{u}_\chi^{\,\mathrm{f}} - \vec{u}_\chi^0 = \frac{2M_\zeta}{M_\chi + M_\zeta}\left(\vec{u}_\zeta^0 - \vec{u}_\chi^0\right)$$

$$\Delta\vec{p}_\chi = \frac{2M_\chi M_\zeta}{M_\chi + M_\zeta}\left(\vec{u}_\zeta^0 - \vec{u}_\chi^0\right) \tag{3.4.4}$$

[4] This trivial solution corresponds to the case where there is no collision, and neither the individual momentums and energies of the particles change

$$\begin{aligned}\Delta\vec{u}_\zeta &= \vec{u}_\zeta^{\,\mathrm{f}} - \vec{u}_\zeta^{\,0} \\ &= \frac{2M_\chi}{M_\chi + M_\zeta}\left(\vec{u}_\chi^{\,0} - \vec{u}_\zeta^{\,0}\right)\end{aligned}$$

$$\Delta\vec{p}_\zeta = \frac{2M_\chi M_\zeta}{M_\chi + M_\zeta}\left(\vec{u}_\chi^{\,0} - \vec{u}_\zeta^{\,0}\right)$$

Notice how $\Delta\vec{p}_\chi = -\Delta\vec{p}_\zeta$, which makes sense seeing how the momentum that one species loses is equal to the momentum the other gains.

Finally, to be able to write an expression for the corresponding force, we need the interval of time in which this change of momentum occurs.
This is precisely the definition of the mean time τ_c [5], the average time that it takes a given species to have a change in momentum $\Delta\vec{p}$ through collisions.
This mean time will also be the same for each species ($\tau_c^\chi = \tau_c^\zeta$), which combined with the fact that the change in momentum is opposite for each species, will give a force which is consistent with Newton's third law.

Finally, putting all of this together and returning to the usual plasma variables (also removing the 0 superscript, now knowing that the velocity $\vec{u}$ would correspond to its current value), the force due to the collisions between the species that make up the plasma will be given like:

$$\boxed{\vec{F}_{\mathrm{C}}^{\chi} = \Delta V \frac{n_\chi n_\zeta m_\chi m_\zeta}{n_\chi m_\chi + n_\zeta m_\zeta}\frac{\vec{u}_\zeta - \vec{u}_\chi}{\tau_c^\chi}}$$

The expression for this force can be simplified for the specific case of a fusion plasma, where the main considered species are the fuel ions i, and the electrons e.
Considering these two species, we would have that $n_i m_i \gg n_e m_e$, and for a quasineutral plasma we would also have that $n_i \approxeq n_e = n$.
Making these changes, the collision force for the ions (which would be the same but opposite for the electrons) could be written like:

$$\vec{F}_{\mathrm{C}}^{i} \approxeq \Delta V n m_e \frac{\vec{u}_e - \vec{u}_i}{\tau_c^i}$$

A derivation of τ_c can be also be made through the cross section of the collisions, and the particles' velocity and number density. This analysis for the collisions between ions and electrons gives the expression:

$$\tau_c^i = \frac{6\sqrt{2}\pi^{3/2}\varepsilon_0^2 m_e^{1/2}}{q_e^4 n \ln(\Lambda)}\left(k_B T\right)^{3/2}$$

[5] The definition of the mean collision time can also be modified so that it absorbs the extra 2 factor in the expression for the force

Where T is the average temperature of the plasma (average between electron and ion temperatures), q_e is the charge magnitude of the electron, and $\ln(\Lambda)$ is the Coulomb logarithm, which has a value of around 20 for a fusion plasma.

3.5 Equation of Momentum Conservation

Now that we have seen the way that the change of momentum is expressed in a fluid, as well as the different forces that cause this change in a fluid made of charged particles, we can write the total equation that will give Newton's second law for a plasma fluid.
If we have a fluid element of volume ΔV, where there are χ species that can interact with each other through Coulomb collisions, then the total force that will act on the species ζ will be given like:

$$\boxed{\begin{aligned}\Delta V n_\zeta m_\zeta \left(\frac{\partial \vec{u}_\zeta}{\partial t} + \left(\vec{u}_\zeta \cdot \vec{\nabla}\right)\vec{u}_\zeta\right) = \Delta V n_\zeta q_\zeta \left(\vec{E} + \vec{u}_\zeta \times \vec{B}\right) - \Delta V \vec{\nabla} P_\zeta \\ + \Delta V \sum_{\chi \neq \zeta} \frac{n_\chi n_\zeta m_\chi m_\zeta}{n_\chi m_\chi + n_\zeta m_\zeta} \frac{\vec{u}_\zeta - \vec{u}_\chi}{\tau_c^\zeta}\end{aligned}} \tag{3.5.1}$$

There is an underlying assumption which makes the use of this equation invalid for certain cases.
In the derivation for the pressure gradient force in Section 3.3, we assumed that the distribution function that described the particles in the plasma was a Maxwell-Boltzmann distribution. In Section 3.4, we also saw how Coulomb collisions were the mechanism through which the particles in the plasma entered thermal equilibrium.

This is to say, the collisions between the particles will be the mechanism through which the particles relax into the Maxwellian distribution, and Equation 3.5.1 will be valid to describe the behaviour of the plasma once this has happened.
It is to be expected that the particles relax into the Maxwellian after a time of around τ_c, so then we conclude that this momentum equation will be valid to describe a fusion plasma and its behaviors with a time scale larger than τ_c.

There are, however, a lot of phenomena in a fusion plasma that have time scales shorter than τ_c. For these, they will alter the distribution of the particles quicker than they can enter thermal equilibrium, and then the real distribution of the particles will no longer be a Maxwell-Boltzmann distribution.
For these phenomena, Equation 3.5.1 will no longer give a totally accurate description of the fusion plasma.

Still, this inadequacy will sometimes not be significant enough to make it completely invalid, and this is why the fluid description, and thus the above equation, is still useful to describe a great part of the behaviors of a fusion plasma. For those cases where the fluid model lacks, it is necessary to use a kinetic model.

Chapter 4

Conservation of Energy

We can also find an equation that gives the conservation of energy for the plasma fluid. This equation will also be given for each species that makes up the plasma.
Just as for the momentum equation, we first need to write the energy of the fluid, and then the way it changes through time.

There will be multiple sources and sinks of energy for a fusion plasma, particularly one that uses D-T fuel. The sources will be:

- Ohmic heating power
- External heating power
- Fusion power through α particles

While the energy losses will be due to:

- Thermal conduction
- Bremsstrahlung radiation

There will also be other terms, which will represent either a gain or a loss in the energy of the plasma (or a given species) depending on the situation. These are:

- Work done by compression or expansion
- Collisions with other species

It's worth noting here that the energy conservation equation that will be obtained at the end is mainly of thermal energy, meaning that it will not take into account the long-range electromagnetic potential energy of the particles [1]. The inclusion of the electromagnetic potentials (due to internal as well as external fields) is easier to make using the kinetic model.

[1] To a certain extent, the short range effects are once again taken into account through the collisions. The bremsstrahlung term also accounts for a fraction of the electromagnetic interactions that can happen between the particles.

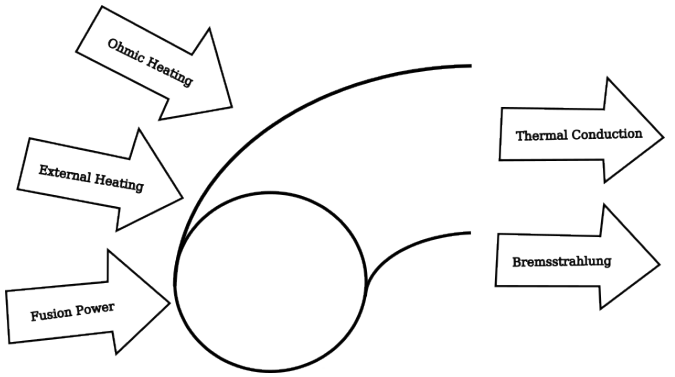


Figure 4.1: Diagram showing the energy sources (arrows pointing in) and sinks (arrows pointing out) of the plasma.

4.1 Change in Internal Energy

The thermal energy of the particles inside of the plasma is due to their thermal motions, meaning due to their thermal velocities $\vec{w}$.
This means that to find an expression for this thermal energy (per particle), we can find the average of the kinetic energy (using the thermal velocities) over the particle distributions, which will once again be treated as a Maxwell-Boltzmann distribution:

$$\begin{aligned} U_{\mathrm{T}} &= \left\langle \frac{1}{2} m w^2 \right\rangle \\ &= \frac{1}{2}\frac{m}{n} \int w^2 f_{\mathrm{M}} \, \mathrm{d}\vec{w} \\ &= \frac{1}{2}\frac{m}{n} \int \left(w_x^2 + w_y^2 + w_z^2\right) f_{\mathrm{M}} \, \mathrm{d}\vec{w} \end{aligned}$$

This is a very similar integral to the one that was calculated back in Section 3.3, with the difference that now we are taking into account all the components of $\vec{w}$ (whereas before we only integrated the $\hat{x}$ component).
The integration over the other two components of the thermal velocity will yield the same result, so the total result in this case will be three times that which was obtained back then [2]. Multiplied by the leading 1/2, this will be:

$$U_{\mathrm{T}} = \frac{3}{2} k_B T$$

We could then multiply this result by $\Delta V n$ to have the total energy of all the particles inside a given fluid element.

[2] Here can be noted the relation between this multiplicative factor of three, and the degrees of freedom of the particles.

What we want is the change through time of this amount, so taking the time derivative and transforming it into the Eulerian coordinate system, it is written like:

$$\boxed{\frac{\mathrm{d}U_{\mathrm{T}}}{\mathrm{d}t} = \frac{3}{2}\Delta V n k_B \left(\frac{\partial T}{\partial t} + \vec{u}\cdot\vec{\nabla}T\right)} \tag{4.1.1}$$

4.2 Ohmic Heating

Ohmic heating refers to the heating of the plasma when a current is induced into it, which is an important component of the functioning of a tokamak reactor. When the current is induced, electrons in the plasma gain speed relative to that of ions (because they have opposite charges), and in their movement they can collide with them. These collisions, or friction between electrons and ions, heats the ions and the plasma overall.

The power provided by the induced electromotive force will heat up the plasma, in a similar way to which a cable heats up when conducting current. The current will be induced into the plasma by action of an (induced) electric field $\vec{E}$, so the force that acts on the particles will be of $\pm q_e\vec{E}$, depending on whether the particles are ions or electrons. Then, if particles move with a velocity $\vec{u}$, the power administered to a given plasma species through this method will be given like $P_\Omega = N\vec{F}\cdot\vec{v} = \pm\left(\Delta V n\right) q_e\vec{E}\cdot\vec{u}$.

Seeing how collisions between plasma species are the cause of ohmic heating, we can start the derivation for the heating power from a simplified version of the equation of momentum conservation, one that only takes into account the electric driving force of the induced current, and the force due to the collisions. Writing these equations for electrons e and ions i, they would look like:

$$\Delta V n m_e \frac{\mathrm{d}\vec{u}_e}{\mathrm{d}t} = -\vec{E}\Delta V n q_e - \Delta V n m_e \frac{(\vec{u}_e - \vec{u}_i)}{\tau_c}$$

$$m_i \frac{\mathrm{d}\vec{u}_i}{\mathrm{d}t} = \vec{E} q_e + m_e \frac{(\vec{u}_e - \vec{u}_i)}{\tau_c}$$

Where $\vec{E}$ can be treated to be constant. The mean time τ_c will also be taken to be a constant.

This is a system of coupled differential equations, which can be solved by writing it in matrix form. Writing $\nu_c = 1/\tau_c$ for ease of writing, this would be:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\vec{u}_e\\ \vec{u}_i\end{pmatrix} = \begin{pmatrix}-\nu_c & \nu_c\\ \dfrac{m_e}{m_i}\nu_c & -\dfrac{m_e}{m_i}\nu_c\end{pmatrix}\begin{pmatrix}\vec{u}_e\\ \vec{u}_i\end{pmatrix} + \begin{pmatrix}-\dfrac{q_e\vec{E}}{m_e}\\ \dfrac{q_e\vec{E}}{m_i}\end{pmatrix}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\vec{U} = M\vec{U} + \vec{b}$$

Here we search for a solution of the type $\vec{U} = \vec{c}\mathrm{e}^{kt}$, where $\vec{c}$ is a vector of constants. We also first try to find the general solution $\vec{U}_\mathrm{G}$, setting $\vec{b} = 0$. With this, we can write:

$$k\vec{c}\mathrm{e}^{kt} = M\vec{c}\mathrm{e}^{kt}$$

$$k\vec{c} = M\vec{c}$$

Which has the form of an eigenvalue equation.

$$(M - \mathbb{I}k)\,\vec{c} = 0$$

$$\det(M - \mathbb{I}k) = 0$$

Following the usual procedure, we first find the eigenvalues k as:

$$\begin{aligned}
\begin{vmatrix} -\nu_c - k & \nu_c \\ \frac{m_e}{m_i}\nu_c & -\frac{m_e}{m_i}\nu_c - k \end{vmatrix} &= 0 \\
(-\nu_c - k)\left(-\frac{m_e}{m_i}\nu_c - k\right) - \frac{m_e}{m_i}\nu_c^2 &= 0 \\
\frac{m_e}{m_i}\nu_c^2 + \nu_c k + \frac{m_e}{m_i}\nu_c k + k^2 - \frac{m_e}{m_i}\nu_c^2 &= 0 \\
k\left(k + \nu_c + \frac{m_e}{m_i}\nu_c\right) &= 0
\end{aligned}$$

$$k = 0,\ -\left(\frac{m_i + m_e}{m_i}\right)\nu_c$$

Now we can find the eigenvectors $\vec{c}$. First, for $k_1 = 0$, we would have:

$$\begin{pmatrix} -\nu_c & \nu_c \\ \frac{m_e}{m_i}\nu_c & -\frac{m_e}{m_i}\nu_c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -\nu_c & \nu_c \\ 0 & 0 \end{pmatrix} =$$

$$c_1^1 = c_1^2$$

$$\vec{c}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $k_2 = (m_i + m_e)/m_i$, the eigenvector would be:

$$\begin{pmatrix} -\nu_c + \dfrac{m_i + m_e}{m_i} & \nu_c \\ \dfrac{m_e}{m_i}\nu_c & -\dfrac{m_e}{m_i}\nu_c + \dfrac{m_i + m_e}{m_i} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \dfrac{m_e}{m_i}\nu_c & \nu_c \\ \dfrac{m_e}{m_i}\nu_c & \nu_c \end{pmatrix} =$$

$$c_2^1 = -\frac{m_i}{m_e} c_2^2$$

$$\vec{c}_2 = \begin{pmatrix} -\dfrac{m_i}{m_e} \\ 1 \end{pmatrix}$$

And with this, we would write the general solution $\vec{U}_{\mathrm{G}}$ as:

$$\begin{aligned} \vec{U}_{\mathrm{G}} &= x_1 \vec{c}_1 \mathrm{e}^{k_1 t} + x_2 \vec{c}_2 \mathrm{e}^{k_2 t} \\ &= x_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \exp(0t) + x_2 \begin{pmatrix} -\dfrac{m_i}{m_e} \\ 1 \end{pmatrix} \exp\left(- \left(\frac{m_i + m_e}{m_i} \right) \nu_c t \right) \end{aligned}$$

We would only be missing the particular solution $\vec{U}_{\mathrm{P}}$:

$$\frac{\mathrm{d}}{\mathrm{d}t} \vec{U}_{\mathrm{P}} = M \vec{U}_{\mathrm{P}} + \vec{b}$$

Seeing how $\vec{b}$ is a constant, we can also look for a solution of constant $\vec{U}_{\mathrm{P}}$. Then, replacing this would give:

$$0 = M \vec{U}_{\mathrm{P}} + \vec{b}$$

Which we can solve like:

$$\begin{pmatrix} -\nu_c & \nu_c \\ \dfrac{m_e}{m_i}\nu_c & -\dfrac{m_e}{m_i}\nu_c \end{pmatrix} = \begin{pmatrix} \dfrac{q_e \vec{E}}{m_e} \\ -\dfrac{q_e \vec{E}}{m_i} \end{pmatrix}$$

$$\begin{pmatrix} -\nu_c & \nu_c \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \dfrac{q_e \vec{E}}{m_e} \\ 0 \end{pmatrix}$$

$$U_{\mathrm{P}}^1 = U_{\mathrm{P}}^2 - \frac{q_e \vec{E}}{m_e \nu_c}$$

$$\vec{U}_{\mathrm{P}} = \begin{pmatrix} 1 - \dfrac{q_e \vec{E}}{m_e \nu_c} \\ 1 \end{pmatrix}$$

And so we would write the total solution like:

$$\vec{U}_{\mathrm{T}} = x_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + x_2 \begin{pmatrix} -\dfrac{m_i}{m_e} \\ 1 \end{pmatrix} \exp\left(-\left(\frac{m_i + m_e}{m_i} \right) \nu_c t \right) + \begin{pmatrix} 1 - \dfrac{q_e \vec{E}}{m_e \nu_c} \\ 1 \end{pmatrix}$$

And finally the individual velocities like:

$$\vec{u}_e = x_1 - x_2 \frac{m_i}{m_e} \exp\left(-\left(\frac{m_i + m_e}{m_i} \right) \nu_c t \right) + 1 - \frac{q_e \vec{E}}{m_e \nu_c}$$

$$\vec{u}_i = x_1 + x_2 \exp\left(-\left(\frac{m_i + m_e}{m_i} \right) \nu_c t \right) + 1$$

The next step would be to solve for the constants x_1 and x_2.
For this, we need to use the initial conditions of the problem. If the electrical field is induced, or "started", at $t = 0$, then we could also assume that the electrons and ions are initially at rest. These conditions would give:

$$\begin{aligned} 0 &= x_1 - x_2 \frac{m_i}{m_e} + 1 - \frac{q_e \vec{E}}{m_e \nu_c} \\ 0 &= x_1 + x_2 + 1 \end{aligned}$$

Subtracting the second equation from the first one, we would find:

$$\begin{aligned} 0 &= x_2 \left(-\frac{m_i}{m_e} - 1 \right) - \frac{q_e \vec{E}}{m_e \nu_c} \\ -\frac{q_e \vec{E}}{m_e \nu_c} &= \left(\frac{m_e + m_i}{m_e} \right) x_2 \end{aligned}$$

$$x_2 = -\frac{q_e \vec{E}}{\nu_c \left(m_e + m_i \right)}$$

And we could also use $x_1 = -x_2 - 1$. So replacing this back into the equations for the

velocities $\vec{u}$, we would get:

$$\begin{aligned}\vec{u}_e &= (-x_2 - 1) - x_2\frac{m_i}{m_e}\mathrm{e}^{k_2 t} + 1 - \frac{q_e\vec{E}}{m_e\nu_c} \\ &= -x_2\frac{m_i}{m_e}\left(\mathrm{e}^{k_2 t} + \frac{m_e}{m_i} + \frac{1}{x_2}\frac{q_e\vec{E}}{m_i\nu_c}\right) \\ &= \left(\frac{m_i}{m_e + m_i}\right)\frac{q_e\vec{E}}{m_e\nu_c}\left(\mathrm{e}^{k_2 t} + \frac{m_e}{m_i} - \frac{m_e + m_i}{m_i}\right) \\ &= \left(\frac{m_i}{m_e + m_i}\right)\frac{q_e\vec{E}}{m_e\nu_c}\left(\exp\left(-\left(\frac{m_i + m_e}{m_i}\right)\nu_c t\right) - 1\right)\end{aligned}$$

$$\begin{aligned}\vec{u}_i &= (-x_2 - 1) + x_2\mathrm{e}^{k_2 t} + 1 \\ &= x_2\left(\mathrm{e}^{k_2 t} - 1\right) \\ &= \frac{q_e\vec{E}}{\nu_c\left(m_e + m_i\right)}\left(1 - \exp\left(-\left(\frac{m_i + m_e}{m_i}\right)\nu_c t\right)\right)\end{aligned}$$

We can further simplify these expressions using the fact that $m_i \gg m_e$ (so that $m_i + m_e \approxeq m_i$). Also, taking the limit $t \to \infty$ to get the steady state value [3], the velocities would be written as:

$$\begin{aligned}\vec{u}_e &\approxeq \frac{q_e\vec{E}}{m_e}\tau_c\left(\mathrm{e}^{-\frac{t}{\tau_c}} - 1\right) \\ &\approxeq -\frac{q_e\vec{E}}{m_e}\tau_c\end{aligned}$$

$$\vec{u}_i \approxeq \frac{q_e\vec{E}}{m_i}\tau_c$$

A couple of things can be noted here. First, the velocity gathered by the particles is equal to the acceleration that the induced electromotive force gives them, times the mean time between collisions with other particles. The particles gather their velocity while being accelerated in the mean time between collisions.
Another thing is that the velocity of the electrons is much greater than that of the ions.

[3] Steady state refers to the state where the plasma has already stabilized, and its properties no longer change with time. The found functions for the velocities have an asymptotic behavior toward $t = \infty$, and that is why that limit is taken.

This can be understood seeing how electrons react more easily to electromagnetic stimuli due to their lower inertia compared to the heavier ions.
For this reason, when electrons collide with ions, they will transfer a small part of their energy to them, and so they will end up with a greater speed. Still, electron-ion collisions will be the mechanism through which the induced energy into the electrons will end up heating up all of the plasma, not only the electrons.
As can be seen from the above equations and reasoning, the heating (increase in speed) of the electrons is initially greater than that of ions by a factor of m_i/m_e, which translates to about four to six thousand times. Thus, it can be concluded that Ohmic heating preferentially heats the electrons, and the heating of the ions can be neglected.

Now we can return to find an expression for P_Ω. However, instead of replacing the expression for the velocity $\vec{u}_e$, we can replace that for $q_e\vec{E}$. This would give:

$$\begin{aligned} P_\Omega &= \Delta V n \left(m_e \frac{\vec{u}_e}{\tau_c} \right) \cdot \vec{u}_e \\ &= \Delta V \frac{n m_e u_e^2}{\tau_c} \end{aligned}$$

And here we could try to write in terms of the current density $\vec{J} \cong -n q_e \vec{u}_e$ [4]. This would then give:

$$P_\Omega = \Delta V \frac{m_e}{q_e^2 n_e \tau_c} J^2$$

So we can finally write the Ohmic heating power as:

$$\boxed{P_\Omega = \Delta V \eta J^2} \tag{4.2.1}$$

With η being the plasma resistivity, also called the Spitzer resistivity:

$$\begin{aligned} \eta &= \frac{m_e}{q_e^2 n_e \tau_c} \\ &= \frac{\sqrt{2} q_e^2 m_e^{1/2} \ln(\Lambda)}{12\pi^{3/2}\varepsilon_0^2} \frac{1}{(k_B T)^{3/2}} \end{aligned} \tag{4.2.2}$$

The main thing that can be noted here is that Equation 4.2.1 is the equivalent of $P = I^2 R$, which gives the power P that is dissipated by a resistance R through which a current I is circulating [5].

[4] $\vec{J}$ is actually defined as equal to $q_e n (\vec{u}_i - \vec{u}_e)$, but the contributions from the ions are ignored because of their high mass (which gives them a small velocity).

[5] Note that in the case of Equation 4.2.1, η has dimensions of electrical resistance times length, while $\vec{J}$ is current over area. Then, this expression gives a quantity with dimensions of current squared times electrical resistance, over volume. Multiplied by the volume of the element, this checks out to the premise that P_Ω gives the heating power.

This is the same way in which this expression can be interpreted. If a current density $\vec{J}$ is induced into a plasma with a certain resistivity η (which is completely determined by the properties of the plasma), then a plasma fluid element of volume ΔV will be heated with a power P_Ω, in the same way that a cable or resistance heats up when current flows through it [6].

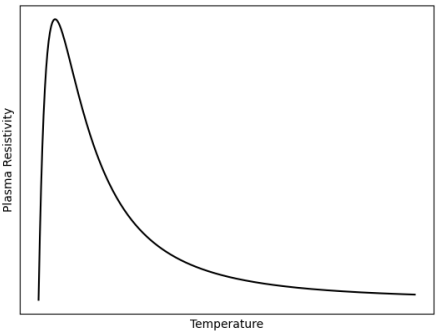


Figure 4.2: Graph showing the trend followed by the Spitzer resistivity, as a function of plasma temperature. For low temperatures, the resistivity rises to a peak, and from there it decreases as temperature increases.

One last thing can be noted here. Looking at Equation 4.2.2 (and its graph in Figure 4.2), it can be seen that the resistivity of the plasma decreases as temperature increases. Seeing how this heating method is directly proportional to the resistivity of the plasma, this means that Ohmic heating will only be an effective heating method up to a certain temperature.
This makes sense, seeing how increasing the temperature of the plasma (or what was initially a gas) increases its grade of ionization. This means that electrons will have a greater energy and they will be less affected by the electromagnetic attraction of the ions. Thus, they will be able to circulate through the apparent ion medium more freely, with less friction, which means a smaller resistivity.
Then, when the plasma reaches a high enough temperature, the value of η will have decreased to a point that current conduction through the plasma will no longer generate a substantial heating power P_Ω.
As it turns out, the heating provided by this method is not enough to heat the plasma up to fusion temperatures. It is for this reason that other heating methods need to be

[6] The power that heats up a resistance is given through this expression because all the power that flows through it will go to heating it. A resistance is just a medium through which a current of electrons encounters a certain grade of opposition, or friction. Then, the energy with which the electrons flow through the resistance will end up turning into friction, or heat, as resistances don't do anything else with that energy. Plasmas, however, do use that energy for other things (like fusion or work). This is taken into account in the balance of sources and sinks of the conservation of energy equation.

implemented into reactor designs in order for them to be able to produce fusion energy.

4.3 External Heating

To heat the plasma up to the required temperatures for fusion, other heating methods are also needed. The mechanisms through which these other methods work are such that they can efficiently provide energy to the plasma (perhaps indefinitely), past a point where inductive methods are no longer practical.
Some examples of these alternative methods are neutral beam injection, in which a beam of neutral particles is used to "push" the plasma particles, giving them more energy. Electromagnetic waves can also be used with this same purpose, as well as increasing the current in the plasma.

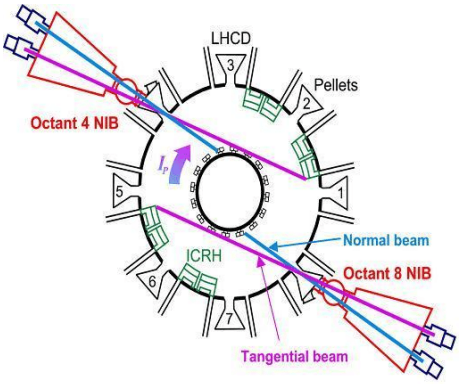


Figure 4.3: Diagram showing the geometry of neutral beam injection. The tangential beam would also increase the plasma's current. Taken from [10].

The specific mechanisms through which each method provides heat and current drive to the plasma require a detailed analysis of the plasma and the interactions it has with each method. For now, we will only consider that each method heats the plasma with a power density $S_{\mathrm{H}}(\vec{r}, t)$, a known function of space and time. This is consistent with the fact that these auxiliary methods can be precisely used to heat the plasma in a specific region and moment.

With this in mind, the other thing that we have to take into account is the fraction of power that is transferred to each species. As we saw in the previous section, some methods heat electrons or ions by different amounts and they transfer a larger part of their power to a specific species.
Even if collisions between the species will ultimately distribute the energy to all the plasma, the fractions r_{H}^{χ} of heating power that goes to each species χ will be important

to determine their power balance.

Then, taking all of this into account, the power P_{H}^{χ} through which a given method will heat the χ species in the plasma (in a fluid element of volume ΔV) will be given by the expression:

$$\boxed{P_{\mathrm{H}}^{\chi} = \Delta V r_{\mathrm{H}}^{\chi} S_{\mathrm{H}}^{\chi}}$$

It is important to keep in mind that for a plasma with k species, the sum of all fractions r_{H}^{k} must be equal to one, as all the power that is absorbed by the plasma has to be distributed between the k species [7].
For example, if we were only considering the electron and ion species of the plasma, we could represent the fraction of the power that goes to the electrons through r_{H}^{e}, and then the fraction of the power that goes to the ions would be given like $1 - r_{\mathrm{H}}^{e}$.

4.4 Fusion Power

The last (assured) energy source term for a fusion plasma fluid would be due to the energy released by fusion reactions.
The derivation of the expression for this type of heating can be first obtained for a general case and fusion reaction, and then made more specific for the case of a D-T (deuterium-tritium) reaction.

First, we can start by finding the density of the rate at which the fusion reactions take place, meaning the number of reactions that happen per unit of time, per unit of volume. If the reaction has a cross section σ, and happens between two atomic species χ and ζ which have a relative velocity $\vec{v} = \vec{v}_{\chi} - \vec{v}_{\zeta}$ between them, then the number of reactions that take place per unit time, per unit volume, is given like:

$$R = n_{\chi} n_{\zeta} \sigma v$$

This expression can be generalized to take into account the whole velocity spectrum of each particle, by replacing the particle densities n by the integral over their respective distribution functions. Then, we would instead have:

$$R = \int \int f_{\chi} f_{\zeta} \sigma \left| v_{\chi} - v_{\zeta} \right| \, \mathrm{d}\vec{v}_{\chi} \, \mathrm{d}\vec{v}_{\zeta}$$

Remembering the definition for an average quantity $\langle Q \rangle$, then the reaction rate (density) can finally be written like:

$$R = n_{\chi} n_{\zeta} \langle \sigma v \rangle$$

[7] From the total power that is output by the heating device, there is also a fraction that is not absorbed by the plasma, but rather reflected by it. However, S_{H} already gives the fraction of the plasma that has indeed been absorbed by the plasma.

And finally, if we multiply the rate R by the energy E_F that is released per fusion reaction, we would obtain the power density S_F generated by fusion reactions like:

$$S_\mathrm{F} = E_\mathrm{F} n_\chi n_\zeta \langle \sigma v \rangle \tag{4.4.1}$$

Now that we have a general expression for the power density generated by fusion reactions, we can make it more specific for the case of a D-T plasma.
First, we can try to find the optimal composition of the plasma. We start from the fact that the plasma is practically charge neutral [8], so the number of electrons will be the same as the number of total ions (remembering that each ion will have one proton). Then, initially ignoring the number of α particles (helium nuclei), this is written as $n_e = n_\mathrm{D} + n_\mathrm{T}$.
This can motivate us to write n_D and n_T as fractions of $n_e = n$, like $n_\mathrm{D} = kn$ and $n_\mathrm{T} = (1-k)\, n$. Then, the optimal fraction between deuterium and tritium atoms in the fuel composition would be the one that maximizes Equation 4.4.1:

$$\begin{aligned}
\frac{\mathrm{d}S_\mathrm{F}}{\mathrm{d}k} &= 0 \\
&= \frac{\mathrm{d}}{\mathrm{d}k} \left(E_\mathrm{F} \left(kn \right) \left((1-k)\, n \right) \langle \sigma v \rangle \right) \\
&= \frac{\mathrm{d}}{\mathrm{d}k} \left(E_\mathrm{F} \langle \sigma v \rangle\, n^2 \left(k - k^2 \right) \right) \\
&= E_\mathrm{F} n^2 \langle \sigma v \rangle \left(1 - 2k \right)
\end{aligned}$$

$$k = \frac{1}{2}$$

And so we find that the optimal composition for the plasma is a 50%-50% mix of deuterium and tritium ions.
Then, the fusion power density would be written like:

$$S_\mathrm{F} = \frac{1}{4} E_\mathrm{F} n^2 \langle \sigma v \rangle \tag{4.4.2}$$

Now, we can finally relate this quantity to the power that heats the plasma. To do this, we have to start by analyzing the products of a D-T fusion, and the way the energy E_F is divided between them.
D-T fusion produces an α particle plus a loose neutron, and the energy E_F will go to their kinetic energies. Seeing how an α particle weights about 4 times more than a neutron, the neutron will end up with four times more kinetic energy than the α particle.

Now, what does this S_F power density have to do with heating the plasma? Given that neutrons are charge neutral, they will not be be confined by the magnetic fields within

[8] This is, if we assume that the initial gas that is ionized to create the plasma is charge neutral.

the plasma. This means that the neutrons will escape the plasma, while the α particles will stay confined and start making part of the plasma.
Then, through collisions, the α particles may transfer their energy to the other plasma species, which are the electrons and the ions. This means that the α particles will be the medium through which the fusion reactions will heat the plasma. Due to the way the total energy is divided between the products, the α particles will transfer only a fifth of the total energy produced in the fusion reaction, while the rest of the energy escapes the plasma with the neutrons.

So, replacing E_{F} with E_α in Equation 4.4.2, we will be able to write the fusion power P_{F} that heats a plasma fluid element like:

$$\boxed{P_{\mathrm{F}} = \Delta V \frac{1}{4} E_\alpha n^2 \langle \sigma v \rangle}$$

Just as in the case of the external heating, the fusion heating power, also called alpha power, will not transfer the power to each plasma species evenly. So for different species χ, this term will be multiplied by a fraction r_{F}^{χ}.

Again, for the case of electrons and ions, we will have a fraction r_{F}^{e} that goes to electrons, and a fraction $1 - r_{\mathrm{F}}^{e}$ that goes to ions.
This fraction is a function of the temperature of the plasma, and for a common temperature of 15 keV, the α particles will transfer about two times more energy to the electrons than to the ions. This will give $r_{\mathrm{F}}^{e} \approxeq 0.3$ [3].

4.5 Thermal Conduction

Now we can start evaluating the terms that represent energy losses for the plasma. While it is empirically known that thermal conduction is one of the most significant of these terms, this is not evident from its initial derivation.
Thermal conduction losses are due to the thermal flow and movement (with a thermal velocity $\vec{w}$) of particles through the boundaries of a fluid element. This makes its derivation very similar to the one for the pressure gradient force (Section 3.3). Depending on the net input or output of particles in the fluid element, thermal conduction would represent an overall kinetic energy gain or loss.

In a similar way to how it was done in Section 3.3, let's first analyze the change in an fluid element's energy due to the flow of particles through the $\hat{x}$ boundaries:

- A particle leaving through the left boundary will decrease the overall energy of the element, as a particle with kinetic energy will be lost.

- A particle leaving through the right boundary will also decrease the overall energy of the fluid element, for the same reason.

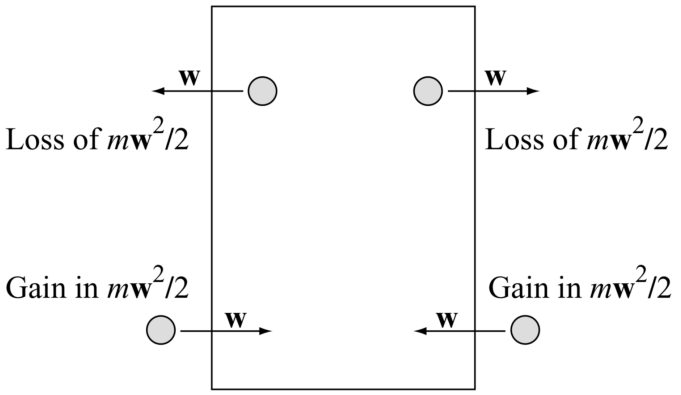


Figure 4.4: Diagram showing the possible particle flows through the boundaries of a fluid element, and their respective change to the element's overall energy. Taken from [3].

- A particle entering through the left boundary will increase the overall energy, because the fluid element will gain a particle with kinetic energy.

- A particle entering through the right boundary will also increase the overall energy of the element, for the same reason.

There are some differences here with what was seen for the momentum change, concerning the change in the quantity due to a particle flow. This is due to the fact that while momentum is a vector (and so can be positive or negative), kinetic energy is a scalar, so any particle flow out of the fluid element will represent a loss in energy, and vice versa.

Now, let's focus on the particles that leave through the right boundary. The corresponding flow of particles will be the same quantity $I_{\mathrm{R}}^{\mathrm{O}}$ (Equation 3.3.1) we had found earlier. This will give the number of particles that exit the right boundary per unit time. Then, to find the change in energy per unit time (power) due to this particle flow, we multiply $I_{\mathrm{R}}^{\mathrm{O}}$ by the kinetic energy of each particle that crosses the boundary (negative because it is an energy loss for the fluid):

$$\begin{aligned} P_{\mathrm{R}}^{\mathrm{O}} &= -I_{\mathrm{R}}^{\mathrm{O}} \left(\frac{1}{2} m w^2 \right) \\ &= -\frac{1}{2} m \Delta y \Delta z \left(\int_0^\infty f_{\mathrm{M}} w^2 w_x \, \mathrm{d}\vec{w} \right) \Big|_{x+\Delta x/2} \end{aligned} \tag{4.5.1}$$

Similarly, for the particles that enter the plasma through the right boundary of the ele-

ment, this integral would read:

$$P_{\mathrm{R}}^{\mathrm{I}} = \frac{1}{2} m \Delta y \Delta z \left(\int_{-\infty}^{0} f_{\mathrm{M}} w^2 w_x \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \tag{4.5.2}$$

Here another difference can be noted with the calculation done with the pressure. Back then, both the entering and exiting of particles changed the element's momentum in the same way, so both of the above integrals had the same sign, and thus could be joined together.
In this case however, particles entering always add to the energy of the element, and particles exiting always subtract. This discrepancy is seen in the opposite signs of the integrals. This also means that both of these effects could cancel each other out, giving no net change in the energy of the fluid element.
Whether they cancel out or not will be dependent on the random velocity of the particles at the given boundary. The net change in energy trough time will then be obtained by integrating the change of energy due to each particle that crosses the boundary ($mw^2/2$), taking into account the entire velocity spectrum of the particles (both positive and negative, meaning both the particles that enter and exit). This would be represented like:

$$\begin{aligned} P_{\mathrm{R}} &= -\frac{1}{2} m \Delta y \Delta z \left(\int_{-\infty}^{\infty} f_{\mathrm{M}} w^2 w_x \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \\ &= -\frac{1}{2} m \Delta y \Delta z \left(\int_{-\infty}^{\infty} f_{\mathrm{M}} \left(w_x^3 + w_x w_y^2 + w_x w_z^2 \right) \, \mathrm{d}\vec{w} \right) \Bigg|_{x+\Delta x/2} \end{aligned} \tag{4.5.3}$$

Where the minus sign is carried over from the fact that the integral was also negative for the values of $w_x > 0$ (the integral from 0 to infinite).

A problem can be spotted in this last form of the integral. A pure Maxwellian distribution f_{M} is an even function on w_x, while the rest of the integrand is an odd function on this same variable. This means that this integral is going to cancel out and be equal to 0.
This is interpreted in the same way as the signs were interpreted for Equations 4.5.1 and 4.5.2. In fact, it is for this same reason that the integral would cancel out. For a pure Maxwellian, the distribution of the particles velocities $\vec{w}$ is entirely symmetrical, meaning that there is no preference for the direction in which the particles travel. This would make it so the same number of particles enter and exit the fluid element, an so it would gain the same amount of energy as it loses, and there would be no net power acting on the element.

Now, this can clearly not be the case, as plasmas actually do lose energy due to conduction with considerable power. This is possible because, in reality, the distribution function f_{M} for the plasma is not a pure, even Maxwellian, but it also has an odd (as in not even) additional term. This would make it so the above integral would not cancel out completely, and it would also not change the obtained result for the pressure gradient

force, as the additional odd term would just cancel out [9].
However, to be able to write this additional term explicitly we would need to make a more precise description of the plasma using the kinetic model. To avoid this, for now we can just be assured that the proper version of Equation 4.5.3 will in fact not be equal to zero, but will rather be equal to the heat flux vector $\vec{q}$. In the current case, it will be equal to the $\hat{x}$ component of this vector.
Even then, to write a complete expression for this heat flux vector, a detailed analysis is needed of the particle transport in the plasma. For now, we will have to carry on pretending like $\vec{q}$ is a known amount. In that case, we would have:

$$-\frac{1}{2}m\Delta y\Delta z\left(\int_{-\infty}^{\infty} f_{\mathrm{M}}w^2w_x\ \mathrm{d}\vec{w}\right)\Bigg|_{x+\Delta x/2} = -\Delta y\Delta z\left(q_x(\vec{r},t)\right)\Big|_{x+\Delta x/2}$$

To evaluate the same at left boundary, we would have something very similar, just with the signs inverted due to the direction of the particle flows being inverted for gain and losses of energy.
For particles entering the fluid element (which requires $w_x > 0$), it would gain energy with a power given like:

$$P_{\mathrm{L}}^{\mathrm{I}} = \frac{1}{2}m\Delta y\Delta z\left(\int_{0}^{\infty} f_{\mathrm{M}}w^2w_x\ \mathrm{d}\vec{w}\right)\Bigg|_{x-\Delta x/2}$$

And for particles exiting, it would lose energy like:

$$P_{\mathrm{L}}^{\mathrm{O}} = -\frac{1}{2}m\Delta y\Delta z\left(\int_{-\infty}^{0} f_{\mathrm{M}}w^2w_x\ \mathrm{d}\vec{w}\right)\Bigg|_{x-\Delta x/2}$$

Then, the integrals would be joined to give as a result:

$$\begin{aligned} P_{\mathrm{L}} &= \frac{1}{2}m\Delta y\Delta z\int_{-\infty}^{\infty} f_{\mathrm{M}}w^2w_x\ \mathrm{d}\vec{w}\Big|_{x-\Delta x/2} \\ &= \Delta y\Delta z\left(q_x(\vec{r},t)\right)\Big|_{x-\Delta x/2} \end{aligned}$$

With the powers evaluated at both boundaries, we can add them together to get the net change in energy (per unit time) for the fluid element (in the $\hat{x}$ direction). This would

[9] In the current heat conduction integral, the additional odd term would leave an additional odd-odd integral which would not be zero. In the pressure gradient integral, the additional odd term would create an additional even-odd integral, which would be equal to zero, leaving the previous result unaltered.

give:

$$P_x = \Delta y \Delta z \left(q_x \Big|_{x-\Delta x/2} - q_x \Big|_{x+\Delta x/2} \right)$$

Now, doing a Taylor expansion of this expression for small values of Δx, we would get:

$$\begin{aligned} P_x &\approxeq \Delta y \Delta z \left(q_x \Big|_{x-\Delta x/2} - q_x \Big|_{x+\Delta x/2} + \Delta x \left(\frac{\partial q_x}{\partial x} \Big|_{x-\Delta x/2} - \frac{\partial q_x}{\partial x} \Big|_{x+\Delta x/2} \right) \right) \\ &\approxeq \Delta y \Delta z \left(0 - \Delta x \left(\frac{\partial q_x}{\partial x} \Big|_{x+\Delta x/2} - \frac{\partial q_x}{\partial x} \Big|_{x-\Delta x/2} \right) \right) \\ &\approxeq -\Delta V \frac{\partial q_x}{\partial x} \end{aligned}$$

Doing the same thing for the other two dimensions, we would obtain equivalent expressions, and adding them all together, we would obtain the expression for the power due to thermal conduction as:

$$\boxed{P_{\mathrm{T}} = -\Delta V \ \vec{\nabla} \cdot \vec{q}}$$

The above equation gives us the power that is lost by a plasma fluid element with a heat conduction vector $\vec{q}$, if we can in some way find it.

4.6 Bremsstrahlung

The other term which will represent a loss of energy for the plasma will be the bremsstrahlung term, which is the loss of energy due to the radiation of the particles within the plasma. In a very similar way to how electrons were the only ones that moved and produced the current for the Ohmic heating term (Section 4.2), in this case it will also only be the electrons that are accelerated and radiate away energy.

The radiation of energy in classical electromagnetism is due to the acceleration of charged particles, and this acceleration will be due to the electromagnetic interaction between the particles of the plasma. The force that causes the acceleration will then be the Coulomb force, which will act during a given distance and time.
Joining all of this together, we will get an expression for the power that is radiated by an electron. Integrating this expression over the velocity distribution of the particles, taking into account all the possible distances and times of action of the force, we would get the plasma's corresponding power output due to bremsstrahlung radiation like [3]:

$$\begin{aligned} P_{\mathrm{B}} &= -\Delta V \left(\sqrt{\frac{2}{9\pi^5}} \right) \left(\frac{\mu_0 q_e^6}{h \varepsilon_0^2 c m_e^{3/2}} \right) Z_i^2 n_i n_e T_e^{1/2} \\ &= \boxed{-\Delta V C_{\mathrm{B}} Z_{\mathrm{eff}} n^2 T_e^{1/2}} \end{aligned}$$

With T_e being the temperature of the electrons specifically.
The change to Z_{eff}, which is the average atomic number of the ionic species i, is made using the fact that that the plasma is in average charge neutral, so $\sum_i Z_i n_i = n_e$ [10]. Then, we would have:

$$Z_{\text{eff}} = \frac{\sum_i Z_i^2 n_i}{\sum_i Z_i n_i}$$
$$= \frac{\sum_i Z_i^2 n_i}{n_e}$$

$$\sum_i Z_i^2 n_i = Z_{\text{eff}} n$$

Again, this term is derived only taking into account the motion of the electrons, while that of the ions is neglected. Then, this term will only be present in the fluid equations of the electrons.

4.7 Thermodynamic Work

The last terms we need to evaluate for the power balance are the ones that can represent either an energy sink or source, depending on the situation.

The first of these terms can be the work that is done by or into the plasma. As we have seen by now, a fluid element has a pressure p, and if the volume of the fluid element changes, so will the energy of the plasma.
If the volume of the fluid element expands, then it will be doing work, and so it will be outputting energy. On the contrary, if the element is compressed and its volume is decreased, work will be done against the plasma, and energy will be input into it.

To start the derivation, we can look at one of the possible cases, as illustrated in Figure 4.5.
Considering first only the $\hat{x}$ axis, we would have a fluid element with boundaries at the points $x_1 = x - \Delta x/2$ and $x_2 = x + \Delta x/2$. At each of these boundaries the particles would move with a velocity $u_x(x_1) - u_x(x)$ and $-(u_x(x_2) - u_x(x))$ respectively [11], where the velocity on the right boundary has a minus sign because it is moving to the left. At the boundaries, the fluid would also have pressure values $p(x_1)$ and $p(x_2)$.

[10] This relation conveys that, in the plasma, the number of positive charges is equal to the number of negative charges. In the specific case of a D-T plasma, Z is always equal to 1, which ends up giving the familiar relation $\sum_i n_i = n_e = n$.

[11] The difference with the velocity $u_x(x)$ at the middle of the fluid element is taken, as it is from this point that the thickness Δx of the element is measured. Then, we would be interested in the difference of velocities on the boundary and this center point.

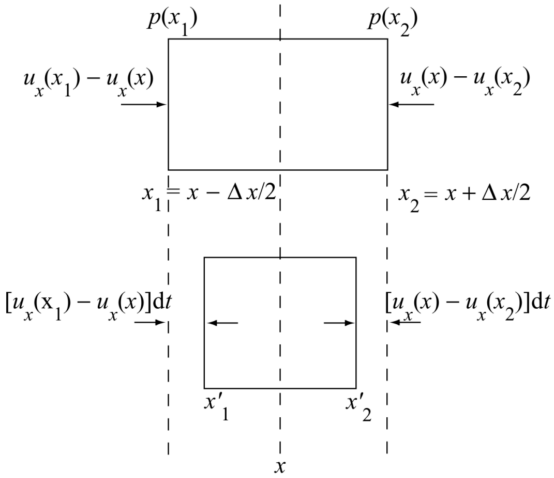


Figure 4.5: Diagram showing how the volume of a fluid element can change depending on the value of the velocities at each of the boundaries. Taken from [3].

A time dt later, the boundaries would have moved to the new points $x_1' = x_1 + (u_x(x_1) - u_x(x))\,\mathrm{d}t$ and $x_2' = x_2 - (u_x(x_2) - u_x(x))\,\mathrm{d}t$, and then the thickness of the fluid element would have changed by quantities $x_1' - x_1 = (u_x(x_1) - u_x(x))\,\mathrm{d}t$ and $x_2' - x_2 = -(u_x(x_2) - u_x(x))\,\mathrm{d}t$.

With the displacement of the boundaries we can then calculate the work done on the element by multiplying by the force acting on each of them, meaning the pressure times the area of the boundary. This would give:

$$\begin{aligned} W_1 &= \Delta y \Delta z\; p(x_1)\,(u_x(x_1) - u_x(x))\,\mathrm{d}t \\ W_2 &= -\Delta y \Delta z\; p(x_2)\,(u_x(x_2) - u_x(x))\,\mathrm{d}t \end{aligned} \tag{4.7.1}$$

The addition of both of these quantities would give the total work done on the fluid element, and dividing this over dt, we would get the power done on the element like:

$$\begin{aligned} P_{\mathrm{W}} &= \frac{W_1 + W_2}{\mathrm{d}t} \\ &= \Delta y \Delta z\,(p(x_1)\,(u_x(x_1) - u_x(x)) - p(x_2)\,(u_x(x_2) - u_x(x))) \end{aligned}$$

To make this expression more concise, we can again Taylor expand for small values of

Δx (which would be equivalent to $x_1 \approxeq x_2 = \tilde{x} \approxeq x$). Then, we would have:

$$\begin{aligned}
P_{\mathrm{W}} &\approxeq \Delta y \Delta z \left(0 + \Delta x \left(\frac{\partial}{\partial x}\left(p(x_1)\left(u_x(x_1) - u_x(x)\right)\right) - \frac{\partial}{\partial x}\left(p(x_2)\left(u_x(x_2) - u_x(x)\right)\right)\right)\right) \\
&\approxeq \Delta V \left(\left(u_x(x_1) - u_x(x)\right)\frac{\partial p(x_1)}{\partial x} + p(x_1)\frac{\partial}{\partial x}\left(u_x(x_1) - u_x(x)\right)\right. \\
&\qquad \left. - \left(u_x(x_2) - u_x(x)\right)\frac{\partial p(x_2)}{\partial x} - p(x_2)\frac{\partial}{\partial x}\left(u_x(x_2) - u_x(x)\right)\right) \\
&\approxeq \Delta V \left(\left(u_x(\tilde{x}) - u_x(x)\right)\frac{\partial p(x_1)}{\partial x} - \left(u_x(\tilde{x}) - u_x(x)\right)\frac{\partial p(x_2)}{\partial x}\right. \\
&\qquad \left. + p(\tilde{x})\left(\frac{\partial u_x(x_1)}{\partial x} - \frac{\partial u_x(x_2)}{\partial x} - \frac{\partial u_x(x)}{\partial x} + \frac{\partial u_x(x)}{\partial x}\right)\right) \\
&\approxeq -\Delta V p(x)\frac{\partial u_x}{\partial x}
\end{aligned}$$

Where multiple terms group and cancel because they are evaluated at $\tilde{x} \approxeq x$. The pressure p, which can also be generalized to the tensor P, is evaluated at the center coordinates of the fluid element.
Finally, generalizing this result to the other two coordinates, the whole term that gives the power input or output by the plasma due to a change in its volume is:

$$\boxed{P_{\mathrm{W}} = -\Delta V P\ \vec{\nabla} \cdot \vec{u}}$$

Two things can be noted here.
First, even though this term has a negative sign, it was seen from the derivation that it can represent a power gain for the plasma. Going back and analyzing Equations 4.7.1, whether the fluid element expands or gets compressed, that is, whether it loses or gains energy, will be dependent on whether $u_x(x_1)$ and $u_x(x_2)$ are greater or smaller than $u_x(x)$. Replacing for values for $u_x(x_1) < u_x(x)$, which would correspond to an expanding volume, we would see that this would give a negative work, or a loss of energy for the plasma. The same would happen for $u_x(x_2) > u_x(x)$.

The second thing that can be noted is the dependence of this term on the pressure of the fluid. There was another term which was dependent on pressure, which was the pressure gradient force. However, this power term is only dependent on pressure, not on its gradient. This means that even if there is no pressure gradient, the plasma may still gain or lose energy through this term.
This then tells us that if there is no force acting on the fluid, then there will not be any change in the velocities of the particles, which means there will be no change in the temperature of the fluid. All of this goes to note how the compression or expansion of the element provides or releases energy into the plasma (or heat Q) in a way that does not change the temperature (related with the internal energy U) through time. Then, we can relate this term with the thermodynamic work W ($p\mathrm{d}V$) term that appears in the first

law of thermodynamics.
Another way in which we can notice this same thing is going back to Equation 4.1.1, and note that even if the temperature of the fluid does not change through time (because there is no force), it can still change though the $\vec{\nabla}T$ term. We would then have a change in temperature, which would represent a change in internal energy U, and along side this work term, we would have $\Delta U = -p\mathrm{d}V$, which can also be recognized as a part of the first law of thermodynamics.

4.8 Collision Exchange

The last term which represents a change in the energy of the plasma is the one that takes into account the energy that is exchanged through collisions [12]. Depending on the energy of each species, they may gain or lose energy through these collisions.
The derivation for this term will follow a similar process to the one made for the collision force. Back then, we found Equations 3.4.2 and 3.4.3, which gave the final average velocities for a collision between plasma species χ and ζ. We can now use this same relations to calculate the change in the kinetic energy of each species:

$$\begin{aligned}\Delta K_\chi &= \frac{1}{2}M_\chi {u_\chi^{\mathrm{f}}}^2 - \frac{1}{2}M_\chi {u_\chi^0}^2 \\ &= \frac{1}{2}M_\chi \left({u_\chi^{\mathrm{f}}}^2 - {u_\chi^0}^2\right) \\ &= M_\chi \left(\vec{u}_\chi^{\mathrm{f}} - \vec{u}_\chi^0\right)\frac{1}{2}\left(\vec{u}_\chi^{\mathrm{f}} + \vec{u}_\chi^0\right)\end{aligned}$$

We can recognize here the product of the first two elements as being equal to the momentum change $\Delta\vec{p}_\chi$ (Equation 3.4.4), while the product between the third and fourth element would be equal to:

$$\begin{aligned}\frac{1}{2}\left(\vec{u}_\chi^{\mathrm{f}} + \vec{u}_\chi^0\right) &= \frac{1}{2}\left(\frac{2M_\zeta}{M_\chi + M_\zeta}\vec{u}_\zeta^0 + \frac{M_\chi - M_\zeta}{M_\chi + M_\zeta}\vec{u}_\chi^0 + \frac{M_\chi + M_\zeta}{M_\chi + M_\zeta}\vec{u}_\chi^0\right) \\ &= \frac{M_\chi\vec{u}_\chi^0 + M_\zeta\vec{u}_\zeta^0}{M_\chi + M_\zeta} \\ &= \vec{V}_{\mathrm{cm}}\end{aligned}$$

Where the last form can be recognized as the velocity of the center of mass $\vec{V}_{\mathrm{cm}}$. Repeating this same process for the other species, we would also get:

$$\begin{aligned}\Delta K_\zeta &= \frac{1}{2}M_\zeta\left({u_\zeta^{\mathrm{f}}}^2 - {u_\zeta^0}^2\right) \\ &= \Delta\vec{p}_\zeta\ \frac{1}{2}\left(\vec{u}_\zeta^{\mathrm{f}} + \vec{u}_\zeta^0\right)\end{aligned}$$

[12] The power that was gained through ohmic heating was also due to friction and collisions between electrons and the rest of the plasma. However, that term is specifically due to currents, that is, for electrons that are being driven by electric fields. This new term seeks to find whatever energy change that is produced due to arbitrary differences in the velocities of the species.

With:

$$\begin{aligned}\frac{1}{2}\left(\vec{u}_\zeta^{\,\mathrm{f}}+\vec{u}_\zeta^{\,0}\right) &= \frac{1}{2}\left(\frac{2M_\chi}{M_\chi+M_\zeta}\vec{u}_\chi^{\,0}+\frac{M_\zeta-M_\chi}{M_\chi+M_\zeta}\vec{u}_\zeta^{\,0}+\frac{M_\chi+M_\zeta}{M_\chi+M_\zeta}\vec{u}_\zeta^{\,0}\right)\\ &= \frac{M_\chi\vec{u}_\chi^{\,0}+M_\zeta\vec{u}_\zeta^{\,0}}{M_\chi+M_\zeta}\\ &= \vec{V}_{\mathrm{cm}}\end{aligned}$$

So then, the change in the kinetic energy energy of each species can be written as:

$$\Delta K_\chi = \Delta\vec{p}_\chi\cdot\vec{V}_{\mathrm{cm}}$$

$$\begin{aligned}\Delta K_\zeta &= \Delta\vec{p}_\zeta\cdot\vec{V}_{\mathrm{cm}}\\ &= -\Delta\vec{p}_\chi\cdot\vec{V}_{\mathrm{cm}}\end{aligned}$$

If this is the change in the energy of the species per collision, then we can divide by the mean time τ_c between collisions to get the change in the power like:

$$\begin{aligned}P_{\mathrm{C}}^{\chi} &= \frac{\Delta K_\chi}{\tau_c^{\chi}}\\ &= \vec{F}_{\mathrm{C}}^{\chi}\cdot\vec{V}_{\mathrm{cm}}\end{aligned}$$

We can carry out the algebra of the dot product to write this power in a more concise way. It can be easier first returning to the change in the kinetic energy, which would be written like:

$$\begin{aligned}\Delta K_\chi &= \frac{2M_\chi M_\zeta}{M_\chi+M_\zeta}\left(\vec{u}_\zeta^{\,0}-\vec{u}_\chi^{\,0}\right)\cdot\left(\frac{M_\chi\vec{u}_\chi^{\,0}+M_\zeta\vec{u}_\zeta^{\,0}}{M_\chi+M_\zeta}\right)\\ &= \frac{2M_\chi M_\zeta}{\left(M_\chi+M_\zeta\right)^2}\left(M_\zeta {u_\zeta^0}^2-M_\chi {u_\chi^0}^2+M_\chi\vec{u}_\zeta^{\,0}\cdot\vec{u}_\chi^{\,0}-M_\zeta\vec{u}_\chi^{\,0}\cdot\vec{u}_\zeta^{\,0}\right)\end{aligned}$$

Now, we can also relate the kinetic energy of each species with their temperature through the relation $m_\chi {u_\chi^0}^2/2 = 3k_BT_\chi/2$, so we would have:

$$\begin{aligned}\Delta K_\chi &= \frac{2M_\chi M_\zeta}{\left(M_\chi+M_\zeta\right)^2}\left(3N_\zeta k_BT_\zeta-3N_\chi k_BT_\chi+M_\chi\vec{u}_\zeta^{\,0}\cdot\vec{u}_\chi^{\,0}-M_\zeta\vec{u}_\chi^{\,0}\cdot\vec{u}_\zeta^{\,0}\right)\\ &= \frac{4M_\chi M_\zeta}{\left(M_\chi+M_\zeta\right)^2}\frac{3}{2}k_B\left(N_\zeta T_\zeta-N_\chi T_\chi\right)+\frac{2M_\chi M_\zeta\left(M_\chi-M_\zeta\right)}{\left(M_\chi+M_\zeta\right)^2}\left(\vec{u}_\chi^{\,0}\cdot\vec{u}_\zeta^{\,0}\right)\end{aligned}$$

Where N_χ is the total number of particles in a fluid element of volume ΔV. Writing each of these terms (and the total masses M_χ) explicitly, and dividing over the mean time τ_c

(which also divides the expressions by 2), the power change in a plasma species χ, due to a collision with another species ζ, would be given like:

$$\boxed{\begin{aligned} P_{\mathrm{C}}^{\chi} &= \Delta V \frac{2 n_\chi n_\zeta m_\chi m_\zeta}{\left(n_\chi m_\chi + n_\zeta m_\zeta\right)^2} \frac{3}{2} k_B \frac{n_\zeta T_\zeta - n_\chi T_\chi}{\tau_c^\chi} \\ &+ \Delta V \frac{n_\chi n_\zeta m_\chi m_\zeta \left(n_\chi m_\chi - n_\zeta m_\zeta\right)}{\left(n_\chi m_\chi + n_\zeta m_\zeta\right)^2} \frac{\vec{u}_\chi^0 \cdot \vec{u}_\zeta^0}{\tau_c^\chi} \end{aligned}}$$

Once again, this expression can be significantly simplified by writing it for the case of electrons and ions in a quasineutral plasma. Making the species χ be the ions i, setting all the particle densities to be equal to n, and using $m_i \gg m_e$, we would have:

$$\begin{aligned} P_{\mathrm{C}}^{i} &= \Delta V n \frac{2 m_i m_e}{\left(m_i + m_e\right)^2} \frac{3}{2} k_B \frac{T_e - T_i}{\tau_c^i} \\ &+ \Delta V n \frac{m_i m_e \left(m_i - m_e\right)}{\left(m_i + m_e\right)^2} \frac{\vec{u}_i^0 \cdot \vec{u}_e^0}{\tau_c^i} \\ &\approxeq \Delta V n \frac{2 m_e}{m_i} \frac{3}{2} k_B \frac{T_e - T_i}{\tau_c^i} + \Delta V n m_e \frac{\vec{u}_i^0 \cdot \vec{u}_e^0}{\tau_c^i} \\ &\approxeq \Delta V n \frac{3}{2} k_B \frac{T_e - T_i}{\tau_{\mathrm{eq}}^i} \end{aligned}$$

Where the last additional term (the one with the velocities) was ignored, and a new mean time τ_{eq} was defined in order to leave this power expression in the form that is usually used.
As we saw, the corresponding power change for the electrons would just be the opposite of this. The energy lost by one species is the same as the other one gains (through these collisions).

One last thing can be noted about this new mean time τ_{eq}. It was defined so that it could "absorb" all the other factors, and present this power term in a more elegant way that just looks like the change in the internal energy U of the plasma fluid:

$$U_i = \Delta V n \frac{3}{2} k_B T_i$$

$$\frac{\mathrm{d}U_i}{\mathrm{d}t} \approxeq \Delta V n \frac{3}{2} k_B \frac{T_e - T_i}{\tau_{\mathrm{eq}}^i}$$

This is similar to the way in which the collision force term is presented as being the change

of momentum:

$$\vec{p_i} = \Delta V n m_i \vec{u}_i$$

$$\frac{\mathrm{d}\vec{p_i}}{\mathrm{d}t} \approx \Delta V n m_e \frac{\vec{u}_e - \vec{u}_i}{\tau_c^i}$$

These equations show how collisions make these quantities change by an average amount ($T_e - T_i$, and $\vec{u}_e - \vec{u}_i$), each time an average amount of time passes (τ_{eq} and τ_c).
In this way, these mean collision times are like decay constants that tell how fast or slow these quantities change with collisions.
Writing the mean time τ_{eq} in terms of the mean time τ_c, we would get $\tau_{eq} = (m_i/2m_e)\,\tau_c$.
With this, we can see that the mean time for energy exchange through collisions is greater than the mean time for momentum exchange, by a factor of m_i/m_e. In other words, two fluid species will end up with the same momentum much quicker than they will end up with the same (thermal) energy.

4.9 Equation of Energy Conservation

Now that we have written expressions for all of the main mechanisms in which the energy of the plasma fluid changes, we can join all of these terms into a single equation that will model how energy changes in a fusion plasma.
Writing this equation in a similar way to which we wrote Equation 3.5.1, if we have χ different species in the plasma, then for a fluid element of volume ΔV, the energy of a species ζ will change like:

$$\boxed{\frac{3}{2}\Delta V n_\zeta k_B \left(\frac{\partial T_\zeta}{\partial t} + \vec{u}_\zeta \cdot \vec{\nabla} T_\zeta\right) = P^\zeta_{\mathrm{Total}}}$$

With P^ζ_{Total} being the total power that changes the energy of the species ζ, given like:

$$\boxed{\begin{aligned} P^\zeta_{\mathrm{Total}} = {} & \Delta V r^\zeta_{\mathrm{H}} S^\zeta_{\mathrm{H}} + \Delta V r^\zeta_{\mathrm{F}} \frac{1}{4} E_{\mathrm{F}} n_\zeta^2 \langle \sigma v \rangle - \Delta V \vec{\nabla} \cdot \vec{q}_\zeta - \Delta V P_\zeta \vec{\nabla} \cdot \vec{u}_\zeta \\ & + \Delta V \sum_{\chi \neq \zeta} \frac{2 n_\zeta n_\chi m_\zeta m_\chi}{\left(n_\zeta m_\zeta + n_\chi m_\chi\right)^2} \left(\frac{3}{2} k_B \frac{n_\chi T_\chi - n_\zeta T_\zeta}{\tau_c^\zeta} + \left(n_\zeta m_\zeta - n_\chi m_\chi\right) \frac{\vec{u}_\zeta \cdot \vec{u}_\chi}{2\tau_c^\zeta}\right) \end{aligned}}$$

If the species ζ referred specifically to electrons e, then the equation would also have the additional terms (but the particle density in the fusion term would still be the ion particle density n_i):

$$\boxed{\frac{3}{2}\Delta V n_e k_B \left(\frac{\partial T_e}{\partial t} + \vec{u}_e \cdot \vec{\nabla} T_e\right) = P^e_{\mathrm{Total}} + \Delta V \eta J^2 - \Delta V C_{\mathrm{B}} Z_{\mathrm{eff}} n_e^2 T_e^{1/2}}$$

As we saw during the derivation of these terms, the left side of the equation represents the change in the internal thermal energy of the plasma fluid element $\Delta U = 3k_B\Delta T/2$. We also saw how the compression and expansion term P_{W} was the equivalent of the thermodynamic work term $-p\mathrm{d}V$.
Then, we can recognize this whole energy conservation equation as being the equivalent to the first law of thermodynamics for this plasma fluid. All the other power terms (besides the work term) would then represent the heat term Q of the general form of this law, that represent the other mechanisms in which the fluid can gain or lose energy.
If Newton's second law (Navier-Stokes equation) was the equation that determined the evolution of the fluid's momentum, then the first law of thermodynamics is the equation that determines the evolution of its energy.

Equations of State

Throughout the derivation of the fluid equations, a pattern may have been noticed. The conservation of particles gave the particle density n as a function of their average velocity $\vec{u}$. The equation for the conservation of momentum gave the average velocity $\vec{u}$ in terms of the fluid pressure p. The relation for the conservation of energy in turn gives the temperature and pressure of the fluid (which are both directly proportional to each other) as a function of the heat conduction in the fluid, represented by the heat conduction vector $\vec{q}$.
We also saw how the derivation of $\vec{q}$ required the use of kinetic theory, and here another pattern can be noticed. The particle density n can be found by integrating the particle distribution f_{M} over the velocity space, while the average velocity $\vec{u}$ is found by integrating f_{M}, times velocity, over the velocity space. Pressure is obtained by integrating f_{M} times the velocity squared, while $\vec{q}$ is found by integrating the distribution times the velocity cubed.

These quantities that are obtained by integrating the distribution function times a certain power of the velocity are called the velocity moments of the distribution, and so we can see how the value of each moment is determined with an equation that is also dependent on the next velocity moment.
This pattern will continue, and the value of the heat flux vector $\vec{q}$, which is necessary to determine the value of the plasma pressure p (which is necessary to determine the value of the average velocity $\vec{u}$, which is ...), needs to be determined through an equation dependent on the next velocity moment of the distribution, and so on.

Then, if we want to be able to determine any quantity of interest in the plasma, we need to find a way of stopping this cycle, which is to find a way of determining one of these plasma parameters without having to use a higher velocity moment.
The way in which this is usually done is by replacing the equation of energy conservation by an equation of state for the plasma, which is able to determine the pressure without having to resort to heat conduction.

These equations of state are relations between thermodynamic parameters of the plasma, so a good place to start could be the equation for the first law of thermodynamics. And if we want to avoid working with heat conduction, we then need to ignore the heat term Q. This would leave us with the equation of an adiabatic process:

$$\begin{aligned} \mathrm{d}U &= -W \\ &= -p\mathrm{d}V \end{aligned} \tag{4.9.1}$$

Of course, as we saw while analyzing the equation of energy conservation in the previous section, this would eliminate all terms from P^{ζ}_{Total} besides the thermodynamic work term. Even if this is a massive oversimplification of the relation we had previously found, it is a necessary one in order to truncate the cycle of velocity moments [13].

Then, we will approximate the plasma to be an adiabatic system, where there is no transfer of energy through heat. Our goal is to derive an equation of state that can determine the pressure p of the fluid in terms of the other lesser velocity moments (particle density n and average velocity $\vec{u}$).
In order to keep using the first law of thermodynamics (Equation 4.9.1), it would be useful to try and write the internal energy U of a fluid element in terms of its pressure. In a general way, the total internal energy for an element with N particles is written like:

$$U = \frac{f}{2} N k_B T \tag{4.9.2}$$

Where f is the number of degrees of freedom of the particles in the plasma (its usual value is 3, for the three dimensions of space). We also know that the pressure of the plasma is given by Equation 3.3.3:

$$p = n k_B T$$

Multiplying this by the volume V of the fluid element, we would obtain the usual equation of state of an ideal gas:

$$\begin{aligned} pV &= nV k_B T \\ &= N k_B T \end{aligned}$$

This is indeed an equation of state that has pressure in it, but it also has temperature, and we would want something simpler and with less parameters. Then, we can continue by

[13] If the only problematic quantity in the equations presented in Section 4.9 is the heat flux vector $\vec{q}$, one might be tempted to just eliminate the term which is dependent on it, as it is only one term from an already very extense equation. This would leave the rest of the equation already determined and able to be used.
However, as was already mentioned, this heat flux term is one of the most, if not the most significant term in the whole equation. So the sole elimination of this term would already be ignoring the most important factor in an equation for the conservation of energy in the plasma fluid. If this is a necessary step in order to continue, we might as well just replace the whole equation that was previously found, with an equation which would be much more simple, which is the case for the equation of state that will be derived.

solving for T in the equation above, and replacing it into Equation 4.9.2 and differentiating, we would get:

$$U = \frac{f}{2} pV$$

$$\begin{aligned} \mathrm{d}U &= \frac{f}{2} \mathrm{d}\left(pV\right) \\ &= \frac{f}{2} \left(V \mathrm{d}p + p \mathrm{d}V\right) \end{aligned}$$

And replacing the above in Equation 4.9.1, we would get:

$$\frac{f}{2} \left(V \mathrm{d}p + p \mathrm{d}V\right) = -p \mathrm{d}V$$

$$\frac{f}{2} V \mathrm{d}p = -\left(1 + \frac{f}{2}\right) p \mathrm{d}V$$

If we divide both sides over $(f/2)\, pV$, we would obtain:

$$\frac{\mathrm{d}p}{p} = -\left(\frac{2+f}{f}\right) \frac{\mathrm{d}V}{V}$$

Here we can change to primed variables, and integrate from a constant value p_0 (V_0 for volume) to a variable value p (V). We would get:

$$\int_{p_0}^{p} \frac{\mathrm{d}p'}{p'} = -\left(\frac{2+f}{f}\right) \int_{V_0}^{V} \frac{\mathrm{d}V'}{V'}$$

$$\ln\left(\frac{p}{p_0}\right) = -\left(\frac{2+f}{f}\right) \ln\left(\frac{V}{V_0}\right)$$

And here we can define the heat capacity ratio γ, also called adiabatic index, put it in the exponent of the logarithm of the volumes, and elevate both sides with an exponential. This would give:

$$\ln\left(\frac{p}{p_0}\right) = -\gamma \ln\left(\frac{V}{V_0}\right)$$

$$\ln\left(\frac{p}{p_0}\right) = \ln\left(\left(\frac{V}{V_0}\right)^{-\gamma}\right)$$

$$\frac{p}{p_0} = \left(\frac{V}{V_0}\right)^{-\gamma}$$

Finally, putting all variables to one side, and all constants to the other, we can obtain the usual equation of state that describes an adiabatic process in an ideal gas:

$$\frac{p}{V^{-\gamma}} = \frac{p_0}{V_0^{-\gamma}}$$

$$pV^{\gamma} = \text{constant}$$

Once again, here we have obtained a simple equation of state that relates the plasma pressure p, this time to the plasma volume.
However, the total plasma volume is not one of the parameters that appears in the previous fluid equations, so we can still improve this state equation by writing it in terms of distribution velocity moments.
We can do this by relating the plasma's volume to its particle density n. If the total number of particles in the plasma N is written like Vn, then we can write $V = N/n$. And finally, if we take N to be constant [14], we obtain the usual form of the equation of state that is used for the species χ of a plasma [4]:

$$\boxed{\frac{p_\chi}{n_\chi^{\gamma}} = \text{constant}} \tag{4.9.3}$$

Where in principle, γ could also take different values for each species.
This equation relates the second velocity moment of the distribution (p) to the zeroth velocity moment (n), and so the recursiveness towards higher velocity moments is broken.

In order to use this equation, the only thing left to determine is the value of γ. This factor has different ways to be defined:

$$\gamma = \frac{2+f}{f} \tag{4.9.4}$$

$$= \frac{C_p}{C_v}$$

Where C_p is the heat capacity at constant pressure, and C_v is the heat capacity at constant volume. These heat capacities give how much heat needs to be added to an object in order to raise its temperature.
For this reason, even if the equation of state was found starting from the adiabatic case where there was no heat transfer, the value of γ is chosen according to the type of heat transfer behavior that is to be modeled.

[14] This may seem like another undesired simplification of the equation of state, but a constant number of particles was also assumed (even if not explicitly) in the previous conservation of energy equation. Even if the fusion energy term implies a change in the number of ions, it does not actually bring a change of this quantity in the equation.

The first of these cases can be that of a plasma compression that happens slowly compared to the speed in which heat is transferred. This would mean that the energy gained by the plasma because of the work that is done on it (or the energy lost because of work done by the plasma) is lost (or gained back) through heat conduction before the temperature of the plasma can change, and the compression would be **isothermal**.
This isothermal compression happens sometimes in plasmas, again, when the compression is slow compared to the speed of heat conduction. This is the case of some waves that compress the plasma along the direction of the magnetic field.
k

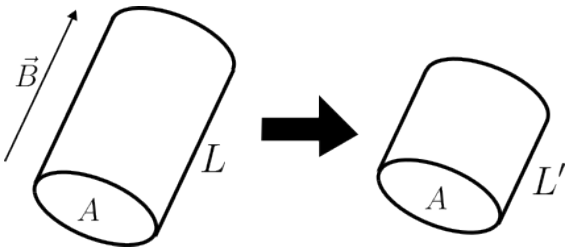


Figure 4.6: Diagram showing the compression of a plasma in the direction parallel to the field.

Parallel to the field, the particles are free to move and travel, and as we saw in Section 4.5, the conduction of heat in the plasma happens due to the flow of particles in and out of the fluid. Then, the compression will not do a lot to prevent the flow of particles, and so the energy will be dissipated as heat before the work from the compression (and expansion) can change the overall energy of the particles, meaning their temperature.
To find the value of γ corresponding to this isothermal compression, we can compare the equation of state (Equation 4.9.3) with the expression that gives the pressure of the plasma (Equation 3.3.3):

$$p = \text{constant } n^{\gamma}$$

$$p = (k_B T)\, n$$

For an isothermal process, the change in volume will change the particle density n, but the plasma's temperature will remain a constant. Then, the quantity inside the parenthesis in the second equation will be a constant, and to make both equations match, we then find that **γ needs to be equal to 1**.
Lastly, going back to Equation 4.9.4, we could notice that this value of $\gamma = 1$ corresponds to the limit where the degrees of freedom f tend to infinity. This does not make much sense, and neither would to consider γ as the ratio of specific heats. The specific heat gives how much energy is needed to generate a change in temperature, but temperature does not change in an isothermal process. However, it does serve to remind us that temperature is not exactly equal to the average energy of particles in a system, but also takes

into account how it distributes among their different degrees of freedom [15].

The next case could be that of a compression that is faster than heat conduction, but slower than energy can be transferred through collisions. Compression faster than heat conduction means that the energy change due to the work will change the energy and temperature of the particles before it can be transferred through heat, so the process will be **adiabatic**.
What **collisions** imply in this case is that the energy will be distributed between all the degrees of freedom of the particles (which will be equal to 3, taking them to be simple monoatomic particles).
The way in which this can be understood is that even if a certain heating method increases the velocity of particles in a certain direction and increases the temperature through a certain degree of freedom [16], random collisions have no way of preserving the original direction of this heating. Then, the heating will actually spread intro the three degrees of freedom.
Three degrees of freedom would give a value of $\boldsymbol{\gamma = 5/3}$, which is the usual case for a monoatomic ideal gas. This is the type of modeling we have been making for the plasma thus far.

The last case is that of a compression which is fast compared to the speed of thermal conduction, and is also fast compared to the rate at which collisions happen. Then, we will have an **adiabatic compression**, and the **lack of collisions** will also mean that the degrees of freedom of the particles will be separated, and the plasma will be treated as anisotropic (with differences depending on the direction).
As we have seen, this would be the most accurate treatment of fusion plasmas, as they tend to be almost collisionless, and the motion tends to be separated into the directions parallel and perpendicular to the existing magnetic field.
Separating the analysis into these same components, we would have one degree of freedom for the parallel direction, which would give a value of $\boldsymbol{\gamma_{\parallel} = 3}$. The perpendicular directions would correspond to the other two degrees of freedom, which would give us a value of $\boldsymbol{\gamma_{\perp} = 2}$.
The fact that the parallel degree of freedom can be counted as being independent of the rest means that if a method can be found to heat this direction specifically, the parallel temperature $T_{\parallel}$ could be increased with special effectiveness, as the supplied energy would stay within this degree of freedom.

Adiabatic Invariants

This separation into the different types of motion of the particles under magnetic fields (parallel and perpendicular to the field) makes it possible to expand the state equations

[15] Supposing the energy of the particles did actually change, if there are infinite degrees of freedom, then energy is divided between all of them and temperature would not increase.

[16] We have to remember here that real plasmas are not isotropic, so pressure (and so to a certain degree temperature) is actually given through tensors, and can be different depending on the direction.

to take into account the magnetic field B (its magnitude) through the use of the adiabatic invariants μ, J and Φ.
The principle of an adiabatic invariant comes from classical mechanics, where for a periodic motion (possibly with slow perturbations compared to the frequency of the motion), the action integral taken over a period gives a constant of the motion. μ would be the adiabatic invariant for the perpendicular motion of the particle, which is a gyration around the lines of the magnetic field (Larmor gyration). J would be the invariant for the motion parallel to the field, which can be periodic for a particle "bouncing" back and forth inside a magnetic mirror. Just as in other fields of physics, these invariants will be very valuable, as they will allow us to find the conditions of the plasma at a later time, if we know its initial conditions.
Starting with the first adiabatic invariant μ, it would be found integrating the perpendicular momentum ($mv_\perp$, with $v_\perp$ being the magnitude of the perpendicular components of the velocity) of the particle over the perpendicular coordinates (parametrized in polar coordinates, with constant radius r_L, corresponding to the Larmor radius of the gyration). This integral would give us the perpendicular invariant $\mathcal{I}_\perp$ like:

$$\begin{aligned}\mathcal{I}_\perp &= \int_0^{2\pi} (mv_\perp)\,(r_\mathrm{L}\mathrm{d}\theta) \\ &= 2\pi m v_\perp r_\mathrm{L} \\ &= 2\pi \frac{m^2 v_\perp^2}{qB} \\ &= 4\pi \frac{m}{q}\mu\end{aligned}$$

Where μ is defined as:

$$\mu = \frac{mv_\perp^2}{2B} \tag{4.9.5}$$

This quantity can be understood more or less easily. If the strength of the magnetic field B (that makes the particles gyrate around it) increases, then the force that makes them gyrate would increase, and so would increase the speed (and kinetic energy) with which the particles gyrate. These quantities (B and the kinetic energy) would increase (or decrease) in a proportion that keeps the value of μ constant.
The second adiabatic invariant J would be found by making the corresponding integral for the parallel motion of the particle in a magnetic mirror. If the particle has turning points a and b, then the invariant $\mathcal{I}_\parallel$ would be found like:

$$\begin{aligned}\mathcal{I}_\parallel &= \int_a^b mv_\parallel \,\mathrm{d}r_\parallel \\ &= mJ\end{aligned}$$

With J being defined as:

$$J = \int_a^b v_{\parallel} \, \mathrm{d}r_{\parallel}$$
$$\approxeq v_{\parallel} L$$

With L being the length of the particle's trajectory, from a to b.
The simplest way of understanding why J should be an invariant is by looking at an example, schematized in Figure 4.7. This shows a diagram of planet Earth, with its magnetic field lines surging from its poles. This field configuration is similar to that of a magnetic mirror, so charged particles can be trapped in the ionosphere, traveling from the north pole to the south pole and back again to the north, represented by the spiral around a magnetic field line. However, due to curvature and gradients of this magnetic

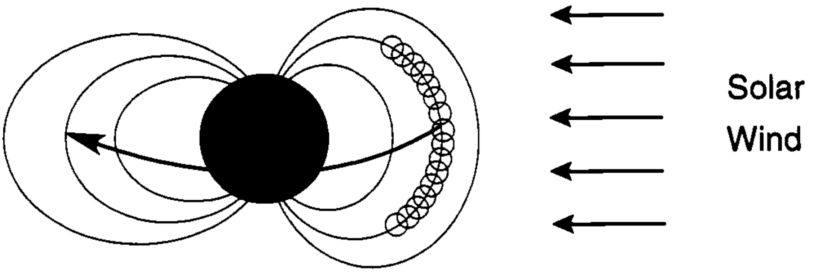


Figure 4.7: Diagram showing the trajectory of a particle trapped in Earth's magnetic field. Taken from [4].

field, the gyration center of the particles also drifts, making the particle also rotate (or precess) around the Earth from east to west, represented by the black arrow that goes from a field line on the right to another field line on the left. On the right of Earth in Figure 4.7, all of the magnetic field lines are on the same longitude (position from east to west), they all lie on the same plane. Each of them, however, are on a different altitude (distance from Earth's surface to the line). Same goes for the field lines to the left.
If the Earth's magnetic field was completely symmetrical, it would make sense for particles to return to the same field line after they have rotated around Earth. After they have returned to their initial longitude, they should be on the same initial altitude. However, as the diagram shows, Earth's magnetic field is distorted by radiation coming from the Sun. This makes the field lines on the right to have different lengths and shapes from the ones on the left. In this case, it would not be obvious to say that particles would still return to the same initial field line once they have completed a precession around Earth. It is here where the invariant J enters. Each magnetic field line in the diagram has a length L and represent a magnetic field strength B different from the others. If the perpendicular kinetic energy of the particle is conserved throughout the motion, Equation 4.9.5 tells us that the strength B of the magnetic field line through which the particle travels will also

remain the same, as the value of μ has to remain constant [17].
Then, if the value of B is unchanged after one precession of the particle around Earth (and in fact, it will remain unchanged during all of the precession), this tells us that the particle will return to the same initial field line once it has returned to the initial longitude. This also tells us that that the particle will eventually return to the original field line of length L, and that this will always be the distance that the particle travels (measured parallel to the magnetic field) when going from a to b. This is what the adiabatic invariant J conveys.
There is one last adiabatic invariant, which is the magnetic flux Φ across the surface created by the drift of the particle. This would be written like:

$$\Phi = BA$$

Where A is the area of the drift surface.
In the situation shown in Figure 4.7, this surface would be the parabolic or round surface enclosed by the precession of the particle around the Earth. The flux of the Earth's magnetic field across this surface would be constant, because as the particle precesses around the planet, it changes altitude in order to keep following lines with the same magnetic field strength B. The flux across the drift surface would be composed of all other lines with a higher strength (because the closer to Earth, the stronger the magnitude of the field).
If there was a change in the radiation incoming from the solar wind (slower than the particle precesses around Earth), then the distortion on the planet's magnetic field would change, and it would be another shape. Still, the particle would alter its original trajectory, to stay at altitudes with constant values of B, and so the magnetic flux across the closed drift surface would remain constant.

Now that we have defined the plasma adiabatic invariants, we can seek to find more precise versions for the equations of state by writing their invariance in terms of the plasma pressure p and particle density n. We can try to do this by remembering the definition $p = mn\left\langle v^2 \right\rangle$ for the plasma pressure from Section 3.3. The set of equations that is obtained is called the Chew-Goldberger-Low (CGL) equations [1].
To start, consider a plasma that is compressed quicker than it can conduct heat, quicker than its rate of collisions, but slower than the Larmor gyration of the particles. In that case, μ would be conserved (remember the mechanical definition of an adiabatic invariant), and we could write:

$$\begin{aligned} \mu &= \text{constant} \\ &= \frac{mv_\perp^2}{2B} \\ &= \frac{p_\perp}{2nB} \end{aligned}$$

[17] In this case, we can say that the value of μ is conserved through the particles motion from north to south because the perpendicular gyration of the particle (related to μ) is faster than this parallel motion. If the change induced by this motion is slower that the Larmor gyration, then it will not be able to affect it substantially, and μ will remain unchanged.

$$\boxed{\frac{p_\perp}{nB} = \text{constant}} \tag{4.9.6}$$

Just as for the previous equation of state, this goes for each species μ in the plasma (the magnetic field would be the one that acts on the species χ).
Let's consider the specific case of a compression in the direction perpendicular to the field [18]. For a compression of this type, the quantity nA would be conserved, because while n would increase (the plasma is compressed, and its length is not changed) , A (the cross sectional area of the plasma) would decrease [19]. The quantity $\Phi = BA$ would also be conserved, because A decreases while B increases [20].

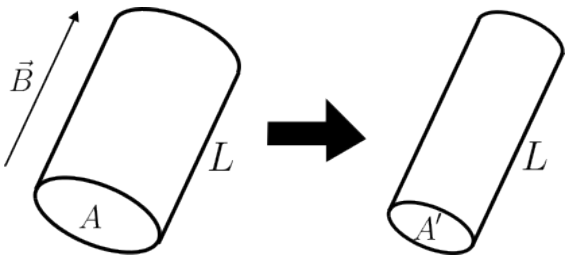


Figure 4.8: Diagram showing the compression of a plasma in the direction perpendicular to the magnetic field.

The conservation of both of these quantities would then tell us that in this case the field B is proportional to the particle density n. Replacing this into Equation 4.9.6, we would recover the regular equation of state, Equation 4.9.3, for the case where $\gamma = 2$, which is the same as we had found previously.

For the parallel motion, we are again going to consider a plasma that is compressed adiabatically, faster than collisions, but slower than particles move back and forth along magnetic field lines. This would ensure that J would remain constant throughout the compression.
To get the state equation for this case, we could start by using the adiabatic invariant Φ. We could write:

$$\begin{aligned}\Phi^2 &= B^2A^2 \\ &= \text{constant}\end{aligned}$$

[18] A compression of this type would be achieved by increasing the strength of the magnetic field. This is because the stronger the field, the smaller the radius r_L of the Larmor gyration, and then the more the plasma particles will bunch up.

[19] This is for a plasma of length ℓ and cross sectional area A, where the total number of particles $N = nV = nA\ell$ is constant. If A is reduced while ℓ remains constant, then n would have to increase in the same proportion as V is decreased, so that N remains constant.

[20] We can recognize this magnetic flux to be the same as the one defined for the adiabatic invariant. If the particles that make up the plasma drift by effect of the magnetic field, we can approximate the cross section of the whole plasma as the area of their drift surface

Then, we could try to introduce the length L of the plasma, and subsequently the invariant $J = v_{\parallel} L$ related to the parallel motion of the particles, by using what would be the total volume $V = AL$ of the plasma:

$$\begin{aligned} \Phi^2 &= \frac{B^2 V^2}{L^2} \\ &= \frac{B^2 V^2 v_{\parallel}^2}{J^2} \end{aligned}$$

Then, if J is a constant of the compression, we would have:

$$B^2 V^2 v_{\parallel}^2 = \text{constant}$$

Next, we could also introduce the particle density n by using the relation $N = nV$, where N corresponds to the total number of particles in the plasma, which is also a constant:

$$\begin{aligned} \frac{B^2 v_{\parallel}^2 N^2}{n^2} &= \text{constant} \\ \frac{B^2 v_{\parallel}^2}{n^2} &= \end{aligned}$$

Finally, we write this in terms of the plasma's parallel pressure $p_{\parallel} = mn \left\langle v_{\parallel}^2 \right\rangle$:

$$\frac{B^2 p_{\parallel}}{m n^3} = \text{constant}$$

$$\boxed{\frac{B^2 p_{\parallel}}{n^3} = \text{constant}}$$

Now lets examine the specific case of a plasma compression in the direction parallel to the magnetic field. This could be achieved without the need of modifying the field B, so it could be treated as a constant too. Then, we would again recover Equation 4.9.3, with a γ value equal to 3, the same we had found earlier.

As we have seen, the use of an equation of state to model the energy evolution in a plasma is an important step to truncate the dependence on the velocity moments of the particle distribution, and to be able to close the system of fluid equations. The use of a state equation is crucial to be able to use the fluid equations to actually model the plasma. The derivation of the first version of the equation of state (Equation 4.9.3) required that all heat flow be neglected in the plasma. Later, it was seen how the heat flow properties were actually taken into account by the quantity γ. Comparing this to what was seen for the equation of energy conservation in Section 4.9, this means we would be simplifying the high descriptiveness of the different power terms into a single scalar quantity (even if the heat flux vector $\vec{q}$ was left undetermined).
While this may seem like a massive downgrade for the description of a plasma, it is not too different from describing plasmas as a fluid. As was mentioned in the beginning of

this chapter, there are many reasons why a real plasma cannot be described as a fluid. Still, the fluid description is capable of modeling a lot of actual plasma behaviors. The situation is similar with the equation of state. While it may not be able to specifically model how energy is transferred between species through collisions, it is still able to model how energy changes after a more general change in the plasma's parameters (pressure and particle density).

The description can also get more precise and accurate by separating the behavior of the plasma into the parallel and perpendicular directions. And with the addition of adiabatic invariants, we can even obtain equations that take into account the magnetic field that acts on the plasma. This is a step towards having an energy equation for the plasma that takes into account the electromagnetic potential that acts on its particles.

Chapter 5

Fusion Plasma Fluid Equations

After having derived equations for the conservation of particles, momentum and energy, we now have a model that is able to describe plasmas as a fluid.
As we have seen, we will have a set of these equations for each species in the plasma, and the equations for one species are dependent on the parameters of the other species' fluid. This makes it so the sets of equations for each species are related and joined into one big total set of equations for the whole plasma.
Proceeding as usual and dividing a fusion plasma into electrons e and fuel ions i, we will have the following set of equations:

Electrons

Conservation of Particles

$$\boxed{\frac{\partial n_e}{\partial t} + \vec{\nabla} \cdot (n_e \vec{u}_e) = 0}$$

Conservation of Momentum

$$\boxed{\frac{\partial \vec{u}_e}{\partial t} + \left(\vec{u}_e \cdot \vec{\nabla}\right) \vec{u}_e = \frac{q_e}{m_e}\left(\vec{E} + \vec{u}_e \times \vec{B}\right) - \frac{1}{n_e m_e}\vec{\nabla} p_e - \frac{\vec{u}_e - \vec{u}_i}{\tau_c^{ei}}}$$

Conservation of Energy

$$\boxed{\begin{aligned}\frac{3}{2} n_e k_B \left(\frac{\partial T_e}{\partial t} + \vec{u}_e \cdot \vec{\nabla} T_e\right) = {} & r_{\mathrm{H}}^e S_{\mathrm{H}}^e + r_{\mathrm{F}}^e \frac{1}{4} E_{\mathrm{F}} n_i^2 \langle \sigma v \rangle - \vec{\nabla} \cdot \vec{q}_e - p_e \vec{\nabla} \cdot \vec{u}_e \\ & - \frac{n_e}{n_i}\frac{3}{2} k_B \frac{n_e T_e - n_i T_i}{\tau_{\mathrm{eq}}^{ei}} + \eta \left(-n_e q_e \vec{u}_e\right)^2 - C_{\mathrm{B}} Z_{\mathrm{eff}} n_i n_e T_e^{1/2}\end{aligned}}$$

(Alternatively) Equation of State

$$\boxed{\frac{p_e}{n_e^\gamma} = \text{constant}}$$

Ions

Conservation of Particles

$$\boxed{\frac{\partial n_i}{\partial t} + \vec{\nabla} \cdot (n_i \vec{u}_i) = 0}$$

Conservation of Momentum

$$\boxed{\frac{\partial \vec{u}_i}{\partial t} + \left(\vec{u}_i \cdot \vec{\nabla}\right) \vec{u}_i = \frac{q_e}{m_i}\left(\vec{E} + \vec{u}_i \times \vec{B}\right) - \frac{1}{n_i m_i}\vec{\nabla} p_i - \frac{n_e m_e}{n_i m_i}\frac{\vec{u}_i - \vec{u}_e}{\tau_c^{ei}}}$$

Conservation of Energy

$$\boxed{\begin{aligned}\frac{3}{2} n_i k_B \left(\frac{\partial T_i}{\partial t} + \vec{u}_i \cdot \vec{\nabla} T_i\right) = r_{\mathrm{H}}^i S_{\mathrm{H}}^i + r_{\mathrm{F}}^i \frac{1}{4} E_{\mathrm{F}} n_i^2 \langle \sigma v \rangle - \vec{\nabla} \cdot \vec{q}_i - p_i \vec{\nabla} \cdot \vec{u}_i \\ - \frac{n_e}{n_i}\frac{3}{2} k_B \frac{n_i T_i - n_e T_e}{\tau_{\mathrm{eq}}^{ei}}\end{aligned}}$$

(Alternatively) Equation of State

$$\boxed{\frac{p_i}{n_i^\gamma} = \text{constant}}$$

Maxwell's Equations

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= \frac{q_e (n_i - n_e)}{\varepsilon_0} \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{B} &= \mu_0 \left(q_e (n_i \vec{u}_i - n_e \vec{u}_e) + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right)\end{aligned}$$

Here it would be good to explain some things about this set of equations. One of the main things to note is how the plasma pressure went back to being a scalar, after having explained that it was actually a tensor. This is done because, this being a system of equations, we have a maximum number of variables that we can find with the limited equations that it represents. Then, using a scalar pressure will allow the system to be closed, rather than having to find all the components of the tensor. This is the same reason to why we use the base version of the equation of state, rather than the anisotropic ones with the adiabatic invariants.
Another thing that can be noted is a change in the expression for the collision term in the conservation of energy from what was shown in Section 4.8. This is because that expression was obtained by approximating quasineutrality ($n_i \cong n_e = n$), while it has not been imposed for this set. The last thing that can be taken into account is that, as we only have two plasma species, the r^χ coefficients that represent the fraction of energy that goes to each species for the heating and fusion terms can be represented as being 1 minus the other species' coefficient. This would be $r^e_{\mathrm{H,\ F}} = 1 - r^i_{\mathrm{H,\ F}}$ [1].

Now, we can go ahead and make an accounting of the total number of equations and variables in this system, to see if it is closed and solvable:

Group	Equations	Unknowns
Conservation of Particles	2 scalar Total: 2	n_e, n_i Total: 2
Conservation of Momentum	2 vector Total: 6	$\vec{u}_e$, $\vec{u}_i$ Total: 6
Conservation of Energy or Equation of State	2 scalar Total: 2 or 2 scalar Total: 2	p_e, p_i Total: 2 or p_e, p_i Total: 2
Maxwell's Equations	2 scalar , 2 vector Total: 8	$\vec{E}$, $\vec{B}$ Total: 6
Total	18 or 18	16 or 16

Table 5.1: Accounting of the total number of equations and unknowns for the fluid system of equations for a fusion plasma.

One precision must be made here for the actual number of equations. For Maxwell's

[1] One other thing which is implicit from these equations is that we are only considering ions with an atomic number $Z = 1$. If it were otherwise, we should have to add a Z variable in a couple expressions, notably those for the plasma charge density ρ and the current J in Maxwell's equations.
However, this would not make much sense, as if we look at the graph of binding energy per nucleon, we are able to see that the maximum energy gain for fusion corresponds for hydrogen fusion, with $Z = 1$. In other words, fusing nuclei other than hydrogen would not make much sense from an energetic standpoint.

equations, we have two scalar equations (the ones with the divergence), and two vector equations, which separated into their 3 different components give us a total of 8 equations. However, the scalar equations are actually redundant, as they can be derived from the other two equations:

$$\begin{aligned}\vec{\nabla}\cdot\left(\vec{\nabla}\times\vec{E}\right) &= 0\\ &= \vec{\nabla}\cdot\left(-\frac{\partial\vec{B}}{\partial t}\right)\\ &= -\frac{\partial}{\partial t}\left(\vec{\nabla}\cdot\vec{B}\right)\end{aligned}$$

$$\vec{\nabla}\cdot\vec{B} = 0$$

And

$$\begin{aligned}\vec{\nabla}\cdot\left(\vec{\nabla}\times\vec{B}\right) &= 0\\ &= \mu_0\vec{\nabla}\cdot\vec{J} + \varepsilon_0\mu_0\vec{\nabla}\cdot\left(\frac{\partial\vec{E}}{\partial t}\right)\\ &= \mu_0\varepsilon_0\left(\frac{1}{\varepsilon_0}\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial t}\left(\vec{\nabla}\cdot\vec{E}\right)\right)\end{aligned}$$

$$\vec{\nabla}\cdot\vec{E} = -\frac{\rho}{\varepsilon_0}$$

Where the fact that the divergence of a curl is always equal to zero[2] was used, as well as the continuity equation for the electric charge. Then, 2 equations will be eliminated from the total count of equations of the system. Still, the total number of equations will be equal to the total number of variables, and so the system will remain closed.
Some other things can be noted here. First, even if the equations for the conservation of energy are partly written in terms of the temperature of the species, the main unknown of these equations is identified as their pressure. This is because the pressure is the next velocity moment, and the previous equation (momentum conservation) is dependent on it. Still, either of the parameters is equivalent, remembering the Equation 3.3.3, which relates the plasma pressure with its temperature.
Also for the equation of energy conservation, we take the heat conduction vector $\vec{q}$ to be known, or at the very least supposable. For the equation of state, the equivalent thing would be to set the value of γ depending on the situation to be modeled. In this way, we could again notice the relation between the adiabatic factor γ and the heat conduction capabilities of the fluid, and we could also recognize another possible reason as to why

[2] This can be understood quickly and intuitively noticing that the curl of a vector will give another vector which will always be perpendicular (because of the cross product) to the gradient operator, so the dot product with it will always be 0.

the equation of state would be preferable over the equation of energy conservation, for an equation that describes the change in the plasma's energy.

As a final comment, we can notice that this system of equations is non-linear, meaning that the change in the quantity to be determined by the equation will not be linear with a linear change in the parameters of the equation. In other words, not every single equation of this system is a common linear equation. This makes it so finding a solution to the current form of the system is a very challenging task.
Depending on the situation that is modeled, some simplifications can be made. For example, for the case of a quasineutral plasma, the particle densities of both species would be the same, so this would effectively eliminate 1 unknown from the system. Also, as we have seen, for a plasma composed of ions and electrons, some terms can be simplified to a great extent by noting that $m_i \gg m_e$. This would also eliminate the collision force term from the equation of momentum conservation for the ions.
However, eliminating some variables and terms for the equations will not offer a solution to the problem of non-linearity. This is solved rather by looking for a specific type of solution close to the equilibrium points of the system, only adding very small perturbations to evaluate the change to the system's solution. If the perturbations are very small, then only the linear (power of 1) terms would remain, and the higher orders would be too small to take into account. This process is called linearization, and it is the main method that can be utilized in order to be able to use these fluid equations in order to model certain plasma characteristics and behaviors.

Chapter 6

Uses of Two-Fluid Equations

Now that we have successfully derived the whole set of two-fluid equations for a plasma fluid, we can try to use them in order to find some solutions for different situations, and thus find some properties and behaviors for the plasma.
The most common use of these equations is actually through the momentum conservation equation, by which one can find some of the particle drifts that can also be found by analyzing the motion of the single particles that compose the plasma. The equation of conservation of energy can also be simplified and converted into the 0-D equation, which can be used to make a power balance for fusion plasmas, and to find the Lawson criterion for ignited plasmas.
As was mentioned, to be able to successfully use these equations, they first have to be simplified into forms that can actually be conveniently solved mathematically. The most common of these modifications is to neglect the effect of collisions between plasma species, as this is a more accurate treatment for fusion plasmas. The other simplifications also tend to take into account the speed or magnitude of certain terms against others, so that the slower and smaller terms can be safely neglected.

6.1 Diamagnetic Drift

One of the possible behaviors which can be studied through the fluid equations for a plasma (and which cannot be done through analyzing single particle motion) is the diamagnetic drift, which consists of a change in the velocity and momentum of the plasma fluid.
To start the derivation, let's first be clear on the type of plasma that will be analyzed. The most important clarification is that we will be considering a fluid without collisions, so there will be no need to take the collision term into account.
The other consideration will be more subtle. We will consider a plasma in which the magnitude of the velocity will change slowly, slower than the rate at which the particles respond to electromagnetic interactions, specially the one that produces the Larmor gyration. This is to say that the electromagnetic force term, specifically the force with the magnetic field $\vec{B}$, will be much greater than the partial time derivative of the veloc-

ity. Then we will be able to omit this term too, and writing the equation of momentum conservation for an arbitrary plasma species, we will have:

$$\left(\vec{u}\cdot\vec{\nabla}\right)\vec{u} = \frac{q}{m}\left(\vec{E}+\vec{u}\times\vec{B}\right) - \frac{1}{nm}\vec{\nabla}p \tag{6.1.1}$$

Now, we wish to evaluate the component of the motion that is perpendicular to the magnetic field. For this, we can change from $\vec{u}$ to only the perpendicular components $\vec{u}_\perp$, and make the vector product with $\vec{B}$. To simplify the expression, we can also note that the parenthesis of the convective term (left had side of the equation) will be equal to a scalar derivative operator, which we will represent as D. Then, we would obtain:

$$\begin{aligned} D\vec{u}_\perp \times \vec{B} &= \frac{q}{m}\left(\vec{E}\times\vec{B} + \left(\vec{u}_\perp\times\vec{B}\right)\times\vec{B}\right) - \frac{1}{nm}\vec{\nabla}p\times\vec{B} \\ &= \frac{q}{m}\left(\vec{E}\times\vec{B} + \vec{B}\left(\vec{B}\cdot\vec{u}_\perp\right) - \vec{u}_\perp\left(\vec{B}\cdot\vec{B}\right)\right) - \frac{1}{nm}\vec{\nabla}p\times\vec{B} \\ &= \frac{q}{m}\left(\vec{E}\times\vec{B} - B^2\vec{u}_\perp\right) - \frac{1}{nm}\vec{\nabla}p\times\vec{B} \end{aligned}$$

Where we used the triple cross product property $\left(\vec{A}\times\vec{B}\right)\times\vec{C} = \vec{B}\left(\vec{C}\times\vec{A}\right) - \vec{A}\left(\vec{B}\times\vec{C}\right)$. We can further simplify this expression if we impose that $D\vec{u}_\perp$ be perpendicular to both $\vec{E}$ and $\vec{\nabla}p$. In this way, the cross product of these vectors would give in different directions (perpendicular directions), and we can also ignore the left hand side of the equation. This same condition requires that the perpendicular component of the velocity be orthogonal also to the electric field and the gradient of the pressure (if D does not change the direction of $\vec{u}_\perp$), and that the electric field $\vec{E}$ and the pressure gradient $\vec{\nabla}p$ be parallel to each other. We will shortly look at a plasma geometry that follows these conditions.

We can then go back to the above equation, and solving for $\vec{u}_\perp$, we can find:

$$\vec{u}_\perp = \frac{\vec{E}\times\vec{B}}{B^2} - \frac{\vec{\nabla}p\times\vec{B}}{nqB^2}$$

As the first term on the right we can recognize a familiar result from the analysis of single particle motion: the $\vec{E}\times\vec{B}$ drift. To the right, we can notice a new drift, which goes in the opposite direction. This is the diamagnetic drift:

$$\boxed{\vec{v}_\mathrm{D} = -\frac{\vec{\nabla}p\times B}{qnB^2}} \tag{6.1.2}$$

We can see a specific example of these drifts in a plasma in Figure 6.1, which shows a cylindrical plasma. In a plasma of this type, the density of particles is higher toward the center of the cylinder [1], so the gradient of the pressure $\vec{\nabla}p$, which points in the same

[1] This is easier to see remembering that a (confined) plasma is closer to a cloud of gas confined to a certain region of space, rather than a liquid with a clear boundary. The density of particles in this cloud decreases away from its center, and somewhere along that line the boundaries of the cloud are defined. Think of something similar to a galaxy, or a cluster of stars.

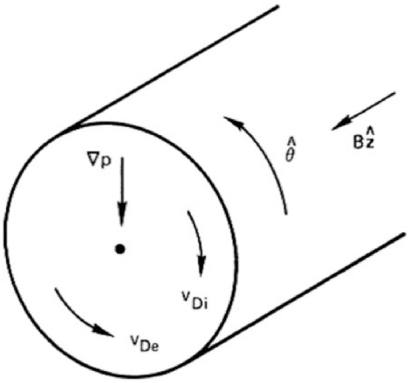


Figure 6.1: Diagram showing the direction of the fields and drifts for a cylindrical plasma. Taken from [2].

direction as the gradient of the particle density $\vec{\nabla} n$ (by Equation 4.9.3, for an isothermal plasma), will point inwards. For this same reason, there will be less charges in the outside of the cylinder, and it is more likely that the electric field $\vec{E}$ will point outwards.

Taking these directions into account, and the resulting direction for the drift, we can notice a couple things. In Figure 6.1, the diamagnetic drifts are labeled v_{Di} and v_{De} for ions and electrons respectively. Both from this diagram, and from Equation 6.1.2, we can notice that the drift will be in opposite directions for opposite charges. This means that this drift will create a current in the plasma, which is called the diamagnetic current:

$$\begin{aligned}
\vec{J}_{\mathrm{D}} &= \sum_{\chi} q_{\chi} n_{\chi} \vec{v}_{\mathrm{D}}^{\chi} \\
&= q_i n_i \vec{v}_{\mathrm{D}}^{\,i} + q_e n_e \vec{v}_{\mathrm{D}}^{\,e} \\
&= q_e n_i \frac{\vec{\nabla} p_e \times \vec{B}}{q_e n_i B^2} - q_e n_e \frac{\vec{\nabla} p_i \times \vec{B}}{-q_e n_e B^2} \\
&= -\frac{\vec{\nabla}\left(p_e + p_i\right) \times \vec{B}}{B^2} \\
&= \boxed{\frac{\vec{B} \times \vec{\nabla}\left(p_i + p_e\right)}{B^2}}
\end{aligned} \tag{6.1.3}$$

Where the calculation was made for a plasma with hydrogen ions, with $Z = 1$.
From here we can notice something else, which is the reason behind the name "diamagnetic". For a plasma with a geometry as the one shown in Figure 6.1, where the pressure gradient is pointing inwards and is perpendicular to the magnetic field, the resulting current will be in a direction such that the magnetic field that it produces points in the

direction opposed to that of the external magnetic field. This is the defying property of diamagnetic objects, and this is the explanation behind the fact that plasmas are diamagnetic.

One last thing can be mentioned about this drift, about its relation to the single particle treatment of the plasmas. Considering just a gradient in the particle density of the plasma, it is not easy to find a reason in the single particle treatment for why a higher density of particles in a certain region would generate a current. Currents in this treatment are generated by opposite drifts of the charges' gyration centers. This is one of the reasons why the diamagnetic drift is not calculated using the single particle treatment. There is no particle gradient drift, at least not in the same way that the field gradient drifts are calculated.
However, we must not forget that the particle density gradient is related to the gradient in the pressure of the plasma, which in turn signifies a force on the particles that compose it. We also know that the single particle treatment predicts that an arbitrary force will generate a guiding center drift. In fact, if we replace the pressure gradient force in the formula that gives the corresponding drift, we get the same opposite drift for ions and electrons, and so we can still predict the diamagnetic current by analyzing the plasma as if it was composed of single particles.

6.2 Bootstrap Current

A very important phenomenon which happens in tokamak reactors is that of the "bootstrap" current, or a plasma current that is generated by the plasma itself, on its own. This increases the plasma current that can be reached in a reactor of this type, which in turn helps to increase it temperature and magnetic confinement.
As it turns out, we can get to understand the bootstrap current to some degree by analyzing a toroidal plasma and trying to find the different types of diamagnetic currents that can arise in it.

When thinking about the basic functioning of a tokamak reactor, we may recognize that the main magnetic fields that are relevant towards the confinement of the plasma inside the vessel are the toroidal and poloidal magnetic fields. The toroidal magnetic fields are generated by the toroidal coils that wrap around the vessel of the reactor, while the poloidal field is created by the electrical current that flows through the plasma.
A toroidal plasma is very similar to a cylindrical plasma, the main difference being only that the torus wraps around the main axis ($\hat{\phi}$ direction), instead of having a given length ($\hat{z}$ direction for the cylinder). An example of a section of a plasma with this geometry can be seen in Figure 6.2.
Then, for a toroidal plasma, we still say that the pressure gradient points from the surface of the torus towards its main axis.

We can then analyze what would be the corresponding diamagnetic current generated by the drift caused by the poloidal magnetic field (we will make an analysis with the

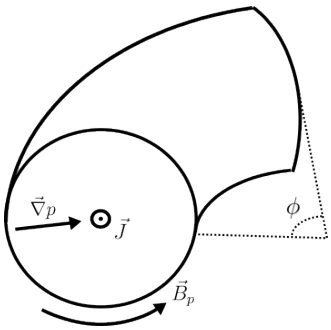


Figure 6.2: Diagram showing a section of a toroidal plasma. It shows the poloidal magnetic field $\vec{B}_p$ generated by the current $\vec{J}$, as well as the plasma pressure gradient $\vec{\nabla}p$.

toroidal field in the next section).
Figure 6.2 shows the direction of this poloidal field. It circles around the plasma, in this case in the anti-clockwise direction given the direction of the current density $\vec{J}$ that flows through the plasma.
To find the corresponding bootstrap current, let us call $\hat{r}_m$ the direction pointing along the minor radius, $\hat{\theta}$ the direction that circles around the minor circle (in the anti-clockwise direction), and $\hat{\phi}$ the direction that points along the major axis (also in the anti-clockwise direction).
Then, $\vec{\nabla}p$ would point in the $-\hat{r}_m$ direction, $\vec{B_p}$ in $\hat{\theta}$ and $\vec{J}$ in $\hat{\phi}$. We could then find the diamagnetic current using Equation 6.1.3:

$$\begin{aligned}\vec{J}_D &= \frac{\left(B_p\ \hat{\theta}\right) \times (-\nabla p\ \hat{r}_m)}{B_p^2} \\ &= \frac{\nabla p}{B_p}\ \hat{\phi} \end{aligned} \tag{6.2.1}$$

And so we find that the diamagnetic current would be pointing in the same direction as the original current $\vec{J}$, meaning that it would be complimenting it and increasing the overall current that flows through the plasma. This can lead us to identify the bootstrap current as a special case of the diamagnetic current in a toroidal plasma.

Now, when investigating deeper into bootstrap current, one may eventually find that the actual theoretical treatment of this phenomenon is made through the study of particle transport in the plasma, taking into account the flux surfaces and the orbits of the particles in the torus. Our previous analysis is an obvious simplification of the fully rigorous study, and while it was useful to qualitatively understand some aspects of the bootstrap current, it is not be very useful to make calculations or to find accurate predictions about related quantities.
Still, if we compare the form we found for the bootstrap current (Equation 6.2.1) with

one we may find from a more advanced analysis (such as the one in chapter 14 of [3], or in chapter 7 of [8]), we can find one big similarity.
As was mentioned, the proper treatment of the bootstrap current is made by analyzing the transport of particles in a toroidal plasma. While these analyses are indeed more advanced than what has been seen in this document, we did analyse particle transport back in Chapters 3 and 4.
More specifically, in Chapter 3, we saw how the transport of particles across a fluid boundary lead to a change in the overall momentum of a fluid element. This change was codified through a force term that was proportional to the gradient of the plasma pressure p.
It is here where we find the similarities between the bootstrap current we found and the others. All of them are also proportional to the pressure gradient $\vec{\nabla}p$. From what was just mentioned, it is no surprise that an analysis of the change in the overall velocities $\vec{u}$ of the particles (which is necessary to be able to find a possible change current in the current density $\vec{J} = qn\vec{u}$ of the plasma) produces an expression containing this $\vec{\nabla}p$ term (given that this pressure gradient force was also found by analysis the transport of particles).
However, we may still ask ourselves why the expression we found is also proportional to $\vec{\nabla}p$, if we never questioned ourselves about the particle transport during its derivation. And while we didn't do so explicitly, we did take into account (to some degree) particle transport implicitly by adding the pressure gradient force to the equation of momentum conservation we initially wrote for finding the diamagnetic drift (Equation 6.1.1). Sometimes, the equation given for the bootstrap current is given instead in terms of the gradient in the plasma's particle density n, but by now we also know of the various ways in which the plasma pressure relates to the particle density ($p = nk_BT$ and the equations of state). So, while our analysis was much simpler than what a phenomenon like the bootstrap current may require, it was still able to show us an important characteristic that defines it.

6.3 Fluid and Single-Particle Treatment

One last thing needs to be clarified for the fluid description of the plasma, as compared to the single particle treatment. In general, the same macroscopic plasma behaviors that can be found through the single particle treatment can also be found through the equations of the fluid treatment. The opposite is also true, with the diamagnetic drift as an example. However, and as we saw, to "port" one result from one treatment to the other, one must be very careful and think closely about the situation and its repercussions in each of the different treatments.

One of the main examples of this is that there is no fluid drift caused by the gradient of the magnetic field perpendicular to it, also called a grad-B drift. The effect that a gradient $\vec{\nabla}B$ would have on a fluid element is that particles in the region where the field is stronger would have a smaller Larmor radius, compared to the radius of gyration of the particles in the region with the weaker field.
However, this would not affect the overall velocity $\vec{u}$ of the fluid, or any other of its properties. The gyration of the particles around the magnetic lines is due to the Lorentz force,

which always has an effect perpendicular to the particles' velocities (ignoring a potential force due to an electric field). Then, this force does not affect the velocity, momentum or energy of the particles, and so it has no it has no effect on a plasma fluid. Figure 6.3

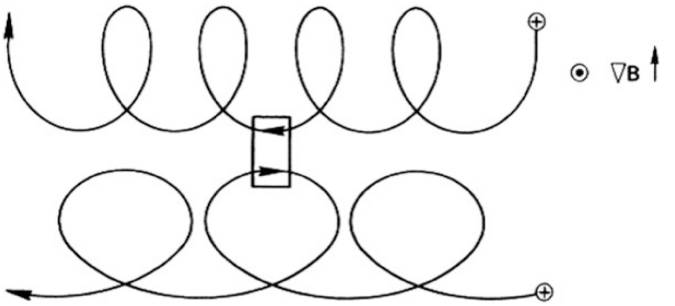


Figure 6.3: Diagram showing another way of understanding why the fluid treatment of plasma has no grad-B drift. Taken from [2].

shows another way of understanding the lack of a grad-B drift in the fluid description. The gradient of the magnetic field would make the particles move to the left, but as can be seen, the contributions of the drift on the top and the bottom of the fluid element cancel each other out. Just as before, this makes it so the grad-B drift has no effect on the fluid element (the fluid element will not drift itself), and so there cannot be an expression for this drift through the fluid treatment.

Still, we know that this drift does exist for the guiding centers in the single particle treatment. Even if it does not change the energy of the particles, it deviates them from their original trajectories. We also know that it has some detrimental effects in the operation of a tokamak fusion reactor. The grad-B drift has opposite directions for electrons and ions in a plasma, and in a toroidal reactor this leads to the concentration of electric charges in the top and bottom of the plasma. This further leads to the generation of a vertical electric field, which introduces instabilities in the plasma due to the $\vec{E} \times \vec{B}$ drift. This is a real effect that has been measured in the past, so the fluid description of the plasma should also be able to predict this same result for the same plasma. To be able to check this, let us first simplify the geometry of the plasma by making it cylindrical instead of toroidal. This will make the calculations much more bearable to do in an analytical way. Then, we can start by calculating the concentration of charge predicted by the single particle treatment.

The main change that will be done to go from the toroidal design of a tokamak reactor to a cylindrical one will be that instead of having a toroidal field that is created by toroidal coils that wrap around the vessel of the reactor, we will have a simpler field $\vec{B}_c$ that goes in the $\hat{\phi}$ direction, which is created by a current I_c that goes through the center of the cylinder, as shown if Figure 6.4. This will produce a similar field to the toroidal one, that will serve the same purpose of guiding the plasma around the reactor.

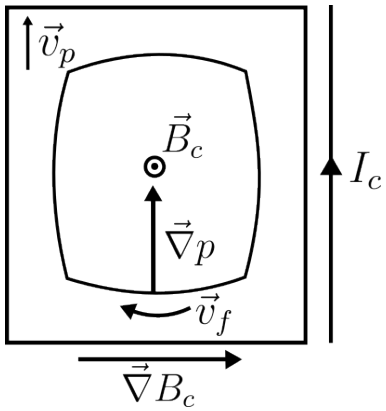


Figure 6.4: Diagram showing a cross section of a plasma in a cylindrical geometry, and the direction of drifts $\vec{v}_p$ and $\vec{v}_f$ predicted by the single particle and fluid treatments, respectively.

To get the correct result, we must also remember that for a magnetic field of the type of $\vec{B}_c$ (no other magnetic field will be modeled), single particle treatment also predicts a drift due to its curvature. So we must also take that curvature drift into account in the calculations.

To start the calculation, we can notice that in this case the gradient of the magnetic field $\vec{\nabla} B_c$ points towards the center of the reactor (See Figure 6.4), seeing how the strength of a magnetic field created by a straight current is proportional to the inverse of the distance. Then, it will be stronger nearer to the center. In this respect, the cylindrical reactor will also be similar to the tokamak, which also has a toroidal field gradient pointing toward the center.
Taking all of this into account, we can write the magnetic field $\vec{B}_c$ and its gradient $\vec{\nabla} B_c$ like:

$$\vec{B}_c \approxeq \frac{B_0}{r}\hat{\phi}$$
$$B_c = \frac{B_0}{r}$$

$$\begin{aligned}\vec{\nabla} B_c &= -\frac{B_0}{r^2}\hat{r} \\ &= -B_c\frac{\vec{r}}{r^2}\end{aligned}$$

Making the appropriate replacement of $\vec{r}$ with $\vec{\nabla} B$ in the expression for the curvature drift, we can write the total drift (grad-B plus the curvature drift) in the gyration center

of the particles due to the toroidal field of a tokamak like:

$$\begin{aligned}\vec{v}_{\mathrm{T}} &= \frac{m\left(v_{\parallel}+v_{\perp}/2\right)}{qB}\frac{\vec{B}\times\vec{\nabla}B}{B^2}\\ &= \frac{k_BT_{\parallel}+k_BT_{\perp}}{qB}\frac{B\hat{\theta}\times(-B\hat{r})}{B^2r}\\ &= \frac{2k_BT}{qrB}\hat{z}\end{aligned}$$

Where we used the equivalence $mv_f^2/2 = fk_BT/2$ between kinetic energy and thermal energy, where f is the degrees of freedom associated to each type of movement. We also took the isotropic case, where the temperature is equal in all directions.
Now that we have calculated the expression for the gyration center drift, proving that it is in the vertical ($\hat{z}$) direction, and opposite for opposite charges, we can calculate the current density $\vec{J}_d$ of charges due to this drift like:

$$\begin{aligned}\vec{J}_d &= nq\left(\vec{v}_T^{\,i}-\vec{v}_T^{\,e}\right)\\ &= \frac{2n\left(k_BT^i+k_BT^e\right)}{rB}\hat{z}\\ &= \frac{2p}{rB}\hat{z}\\ \\ &= \frac{2p}{B_0}\hat{z}\end{aligned}$$

Where we also assumed quasineutrality, and we used the fact that the sum of the ion and electron pressures gives the total pressure p of the plasma.
Ignoring the vector nature of the current density, its magnitude will give us the rate over time at which electric charge accumulates in a surface area in the top of the reactor [2].

$$\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{2p}{B_0}$$

Now, for the fluid treatment of this same situation, we need to take into account the pressure gradient of the plasma. As can be seen in Figure 6.4, we will take a plasma that is denser in its center, so the pressure gradient will point from the edges of the plasma towards that direction. We will also take this pressure gradient to be constant in value all around the plasma.
Then, the pressure gradient will generate a diamagnetic current $\vec{J}_D$ that will circulate around the plasma, going along its edges (clockwise for the case illustrated in Figure 6.4).

[2] The current gives the net flow of positive charges upwards. This is actually the positive charge left by the flow of the negative electrons to the bottom. There, a same but opposite charge area density σ will build up.

Using Equation 6.1.3, we can write the diamagnetic current like:

$$\begin{aligned}\vec{J}_D &= \frac{\vec{B}\times\vec{\nabla}p}{B^2}\\ &= \frac{B\hat{\phi}\times\nabla p\,(-\hat{r},\hat{z},\hat{r},-\hat{z})}{B^2}\\ &= \frac{\nabla p\,(\hat{z},-\hat{r},-\hat{z},\hat{r})}{B}\end{aligned}$$

Again, this current circulates around the plasma, so it is not at first clear why it would lead to a net concentration of charge in the top and bottom of the plasma. If it did, it could be found using the continuity equation for the electric charge, which would in turn require the current density $\vec{J}_D$ to have a divergence.
To check if it does have a divergence, we also need to replace for the correct expressions of the magnetic field's magnitude B at each edge of the plasma. For the vertical edges, we would have constant values, as they are at a fixed distance from the current source that creates the magnetic field. Replacing for these constant values B_k, the vertical components of the current would be:

$$\begin{aligned}\vec{J}_D^{\,\mathrm{v}} &= \frac{\nabla p}{B_k}\,(\hat{z},-\hat{z})\\ &= r_k\frac{\nabla p}{B_0}\,(\hat{z},-\hat{z})\end{aligned}$$

Which would have a divergence of 0 (because they are constant).
For the horizontal edges, the magnetic field would change as a function of r, the distance to the current I_c. The corresponding components of the diamagnetic current would be written like:

$$\begin{aligned}\vec{J}_D^{\,\mathrm{h}} &= r\frac{\nabla p}{B_0}\,(-\hat{r},\hat{r})\\ &= \mp r\frac{\mathrm{d}_z p}{B_0}\hat{r}\end{aligned}$$

Where we replaced ∇p for the corresponding derivative $\mathrm{d}_z = \mathrm{d}/\mathrm{d}z$ (to get the component of the current in the $\hat{r}$ directions, we would need the pressure gradient in the $\hat{z}$ directions). The minus sign corresponds to the top of the plasma, and the plus to the bottom.
This component is not a constant, so we can calculate the only component of the divergence like:

$$\begin{aligned}\vec{\nabla}\cdot\vec{J}_D &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\left(\mp r\frac{\mathrm{d}_z p}{B_0}\right)\right)\\ &= \mp\frac{2\mathrm{d}_z p}{B_0}\end{aligned}$$

And we can relate this to the rate of charge in the charge density of the plasma through

the equation of charge continuity:

$$\begin{aligned}\frac{\mathrm{d}\rho}{\mathrm{d}t} &= -\vec{\nabla}\cdot\vec{J}_D \\ &= \pm\frac{2\mathrm{d}_z p}{B_0}\end{aligned}$$

Now, this result is for the volumetric charge density of the plasma. If we are interested in seeing how the surface charge density changes at the top and bottom, we can integrate this quantity over the vertical ($\hat{z}$) direction, with which we would finally obtain:

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}t} &= \int \frac{\mathrm{d}\rho}{\mathrm{d}t}\,\mathrm{d}z \\ &= \int \pm\frac{2\mathrm{d}_z p}{B_0}\,\mathrm{d}z \\ &= \pm\frac{2p}{B_0}\end{aligned}$$

And so we find that both the single particle and fluid treatments predict the same charge concentration on the top and bottom of the plasma.

While this analysis of a cylindrical plasma constituted a simplification of a more real system with a toroidal geometry, one should expect to obtain similar results when analyzing the same situation for a tokamak reactor, as both geometries produce a very similar situation Figure 6.5 shows the analogous of the cylindrical plasma, but for the case of a

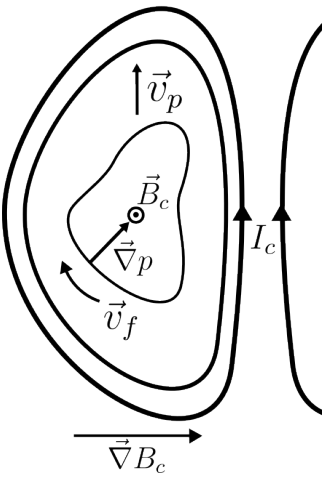


Figure 6.5: Diagram showing a (cross section of a) plasma in a toroidal geometry, similar to that obtained in a tokamak reactor. The corresponding particle and fluid drifts, $\vec{v}_p$ and $\vec{v}_f$ are also labeled.

tokamak reactor. Now, instead of having a single central current that creates the magnetic field $\vec{B}_c$, the field will be generated by the toroidal coils that wrap around the vessel, which will be conducting a current I_c.
Even if this coil configuration is drastically different from the one of the cylindrical reactor, it can still produce a magnetic field of a similar magnitude, pointing in the same $\hat{\phi}$ direction, and with a gradient $\vec{\nabla}B_c$ also pointing in the same direction (towards the inside of the reactor), as the density of toroidal coils will be greater toward the center of the reactor, making the strength of the magnetic field greater in that area.
For the plasma, even if the geometry will also be very different from the previous case, it will still have a pressure gradient that points toward the center of the plasma.
And lastly, given all these similarities, we will also obtain a similar set of particle and fluid drifts. For the case of positively-charged ions, we will have a particle drift $\vec{v}_p$ that will push them toward the top of the reactor. For the fluid drift $\vec{v}_f$, we will obtain an ion drift that makes the particles circulate around the reactor clockwise, in the same direction as for the cylindrical case.
So then, if we are able to work through all the mathematical difficulties that this toroidal geometry conveys, it should not be a surprise to find that not only charge is accumulated at the top and bottom of the vessel, but also that the particle and fluid treatment predict the same charge density in these edges.

The last example we can look at of similar results we can get from both fluid and single particle treatments is that of the bootstrap current. In this case, obtaining the same result will be simple, provided we stay within the simple description of the phenomenon.
For usual plasmas, while the pressure gradient points toward the center of the plasma (the pressure increases closer to the center), the pressure itself will be pointing outwards.
Then, the force acting on particles due to this pressure will also be pointing outwards.
With the direction of this force, we can then find that the resulting particle drift in the poloidal magnetic field will also be pointing in the same direction as the original plasma current that generates the poloidal field. The drift for negative charges will have the opposite direction, and so we find that the force drift is generating a current that compliments the already existing plasma current, which is what defines the bootstrap current.

From what we have seen, it is now clear that both the single particle and the fluid treatment for the plasma can predict the same physical outcome, which comes as no surprise given that both of them are physically valid descriptions for the same system.
Another way in which we can understand the equivalence between both treatments is that, to some extent, the fluid treatment is also a particle treatment. During the introduction, we saw how coherence in a fluid makes it so a fluid element with N particles, each of mass m, can be seen as a "super-particle" of mass $M = Nm$, with a velocity $\vec{u}$ equal to the average of the velocities of each particle in the element. Having this in mind, we can see another degree of equivalence between fluid and particle treatment.
Still, we must remember that plasma fluids are not completely coherent, and this could be what leads to the treatments not being completely equal.

We have also seen how despite this equivalence, it is not always straightforward to get

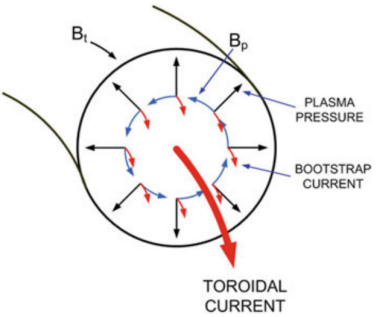


Figure 6.6: Diagram showing the direction of the bootstrap current, as obtained from the single particle analysis, with B_p being the poloidal magnetic field. Taken from [2].

the same result from both methods. For simple descriptions, like the one given for the bootstrap current, it is easy to get the same result. For other phenomena, like the charge concentration we evaluated, or the other drifts derived in chapter 7 of [4], this procedure can be more mathematically involved.
Whatever the case, it is good to be able to know that we can model a plasma as a fluid, and that through this description, for most cases we will be able to find results that help us better understand the plasma under different circumstances.

Bibliography

[1] Michael Brown. *Some notes on MHD equations of state (Chew Goldberger Low)*. Swarthmore College, Unespecified Year.

[2] Francis Chen. *Introduction to Plasma Physics and Controlled Fusion*. Springer International Publishing, 2016.

[3] Jeffrey Freidberg. *Plasma Physics and Fusion Energy*. Cambridge Universtity Press, 2007.

[4] Robert Goldston. *Introduction to Plasma Physics*. IOP Publishing, 1995.

[5] Juan González. *Plasma Current Simulation*. 2026. URL: https://github.com/juandresgg/plasma-current-sim.

[6] Juan Andrés González Vallejo. *Magnetic Confinement Fusion in Tokamak Reactors*. Tech. rep. Universidad de los Andes, 2026. URL: https://hdl.handle.net/1992/79697.

[7] Ian Hutchinson. *Class notes to Introduction To Plasma Physics I*. Fall 2003.

[8] Edward Morse. *Nuclear Fusion*. Springer, 2018.

[9] Wikipedia contributors. *Lists of integrals — Wikipedia, The Free Encyclopedia*. 2026. URL: https://en.wikipedia.org/w/index.php?title=Lists_of_integrals\&oldid=1371952357.

[10] V. Yavorskij et al. "THW/P7-17 Interpretive modelling of neutral particle fluxes generated by NBI ions in JET". In: ().